

AI Clones and Creations

Annie Liang
(Northwestern)

ESIF AI+ML

Introduction

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- Outputs can be rich and complex—**cartoons**

Generating New Yorker-Style Cartoons with AI

Harnessing the power of OpenAI's ChatGPT to Transform Cartoon ideas into Reality



Georgiy Ermakov

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5 min read · Oct 25, 2023

Introduction

Generative AI systems are increasingly used to produce creative content.

Why this is qualitatively new:

- Generation is fast, low cost, and requires little expertise.
- Outputs can be rich and complex—cartoons, **novels**

The AI-generated Books Trend is Getting Worse

It has gotten so bad that Amazon's new policy limits self-published Kindle ebooks to three a day.



Sascha Brodsky, Contributor
January 24, 2024

🕒 4 Min Read

AI • BUSINESS

Introduction

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- Generation is fast, low cost, and requires little expertise.
- Outputs can be rich and complex—cartoons, novels, **actors**

Tilly Norwood: how scared should we be of the viral AI ‘actor’?

A bunch of code is being pushed as the next Scarlett Johansson, a creation that is already causing pushback from real human actors

Tilly Norwood



A 2025 AI-generated image of Tilly Norwood, taken from the character's [Instagram](#) account

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**AI Agents Simulate 1,052 Individuals'
Personalities with Impressive
Accuracy**

Stanford HCAI, 2025

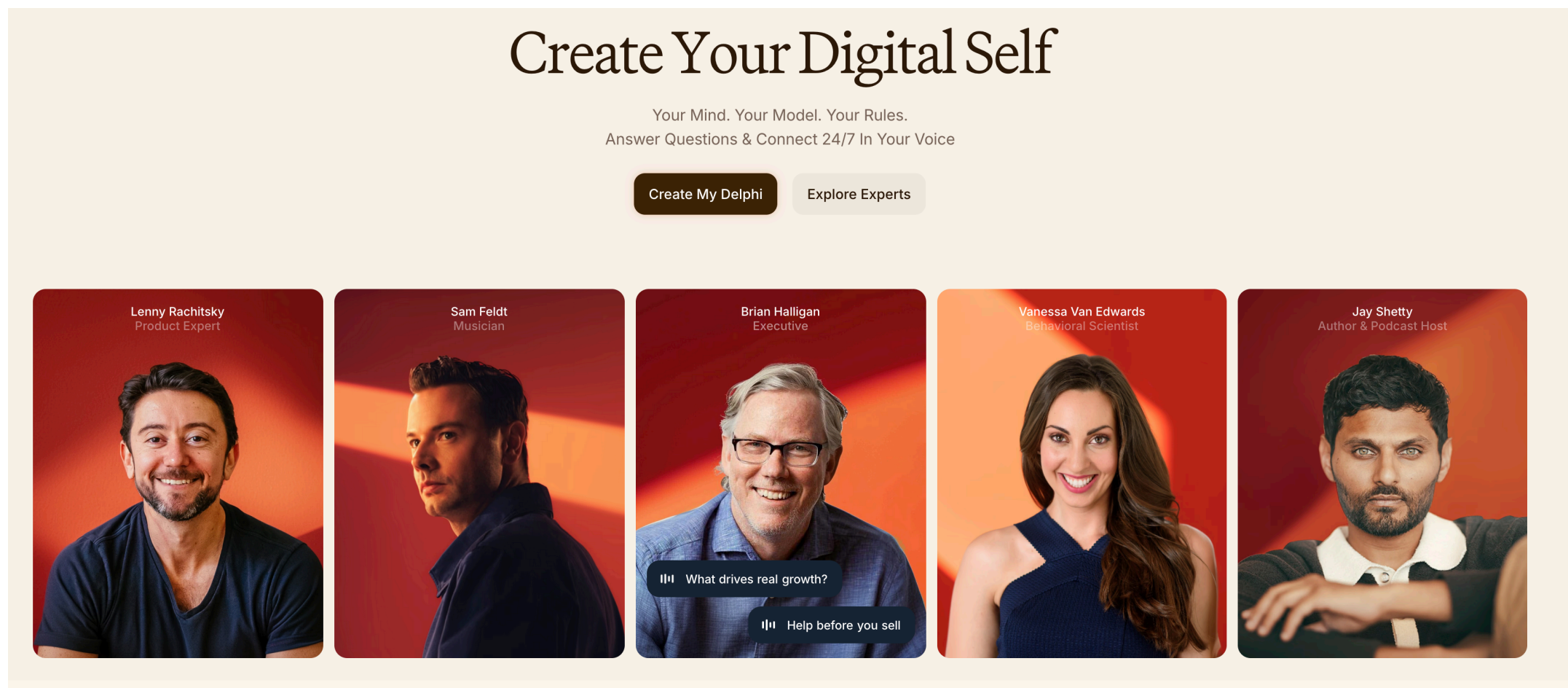
It's no longer science fiction: Your personality – your beliefs, quirks, and decision-making patterns – can be captured and brought to life inside an artificial mind.

Introduction

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Introduction

AI clones and creations will change markets, reshaping

- how people interact with one another
- what and how humans create

We need to think about their consequences and whether/how to regulate them, e.g., how to allocate creative ownership and rights.

These issues are **already** present given existing AI technology, and will grow even more so as this technology improves.

This Talk

Will discuss two aspects of this through two papers:

- Artificial Intelligence Clones

What are the welfare implications of the use of AI clones for searching/matching over human candidates?

- Creative Ownership in the Age of AI (joint with Jay Lu)

How should we define and protect intellectual property in a post-generative AI world?

Artificial Intelligence Clones

Annie Liang
(Northwestern)

Introduction

LLMs, fine-tuned on our personal data, may soon be able to mimic individual personalities (“AI clones,” “AI agents,” “AI delegates”)

This technology has the potential to transform how we search over human candidates, in particular in settings where direct evaluation is costly.

Labor and dating markets are already moving in this direction.

REPLICANT

AI-powered candidate replication platform helping job seekers create intelligent digital representatives for recruitment conversations.



AI Training

Upload your resume and provide context about your experience, goals, and preferences



Clone Creation

Our AI creates an intelligent representation that understands your professional profile



Smart Matching

Your AI clone engages with recruiters, answering questions and showcasing your skills

Ready to Get Started?

Join the future of recruitment and let your AI clone work for you.

Create Your AI Clone

Ribbon raised \$8.2M to fix hiring globally >

Meet your AI-powered recruiter

Ribbon talks with every candidate, scores their answers, and shares insights that help you hire faster and smarter.

Book a demo

Start for free



Daphne D.
Operations Manager

This was my first time doing an interview like this and it was so smooth!



Hassan J.
DevOps Engineer

I like the new AI approach, and I thought the agent was very interactive and smart and responded well to changing dynamics of the conversation!



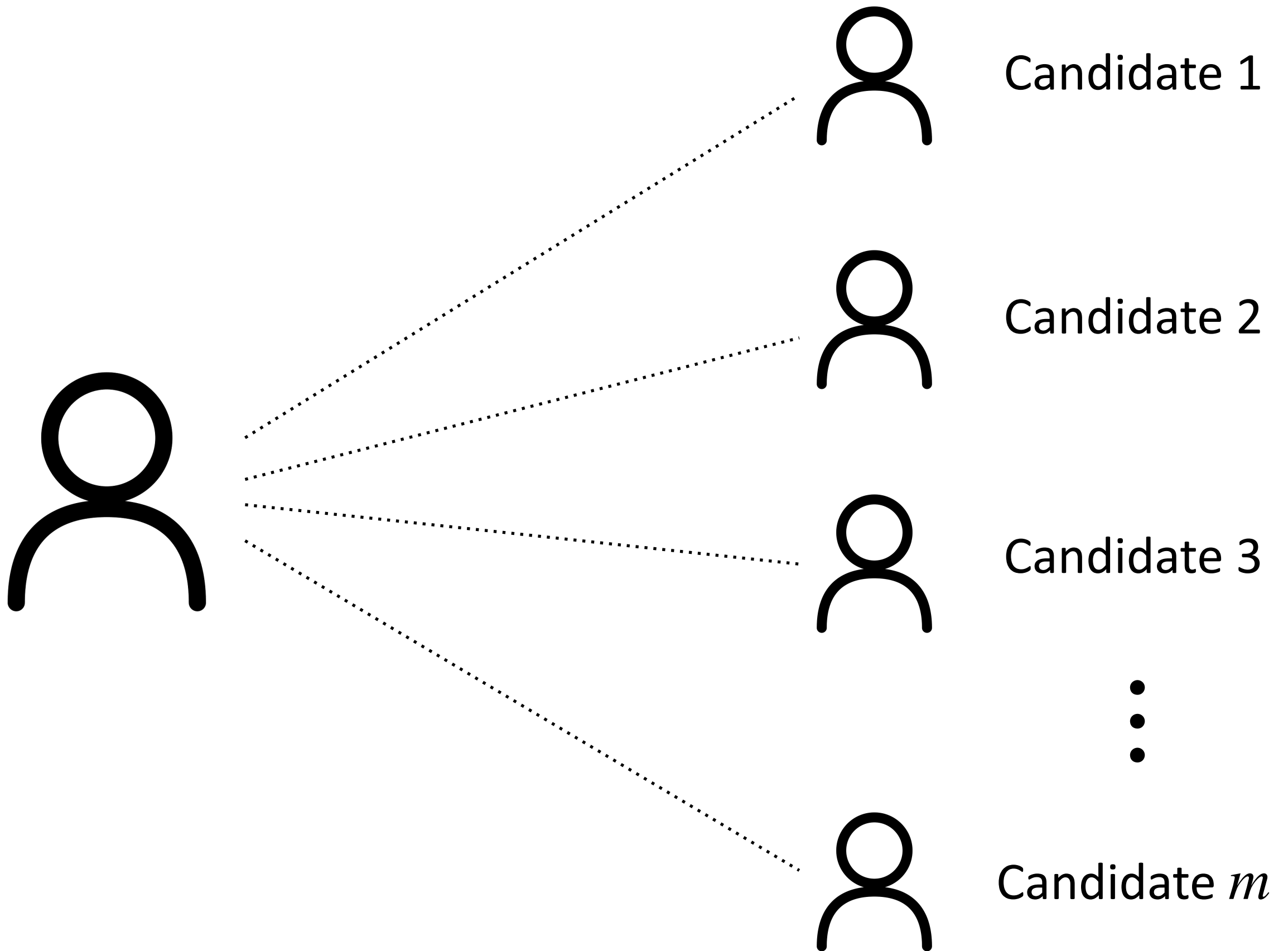
Jamie K.

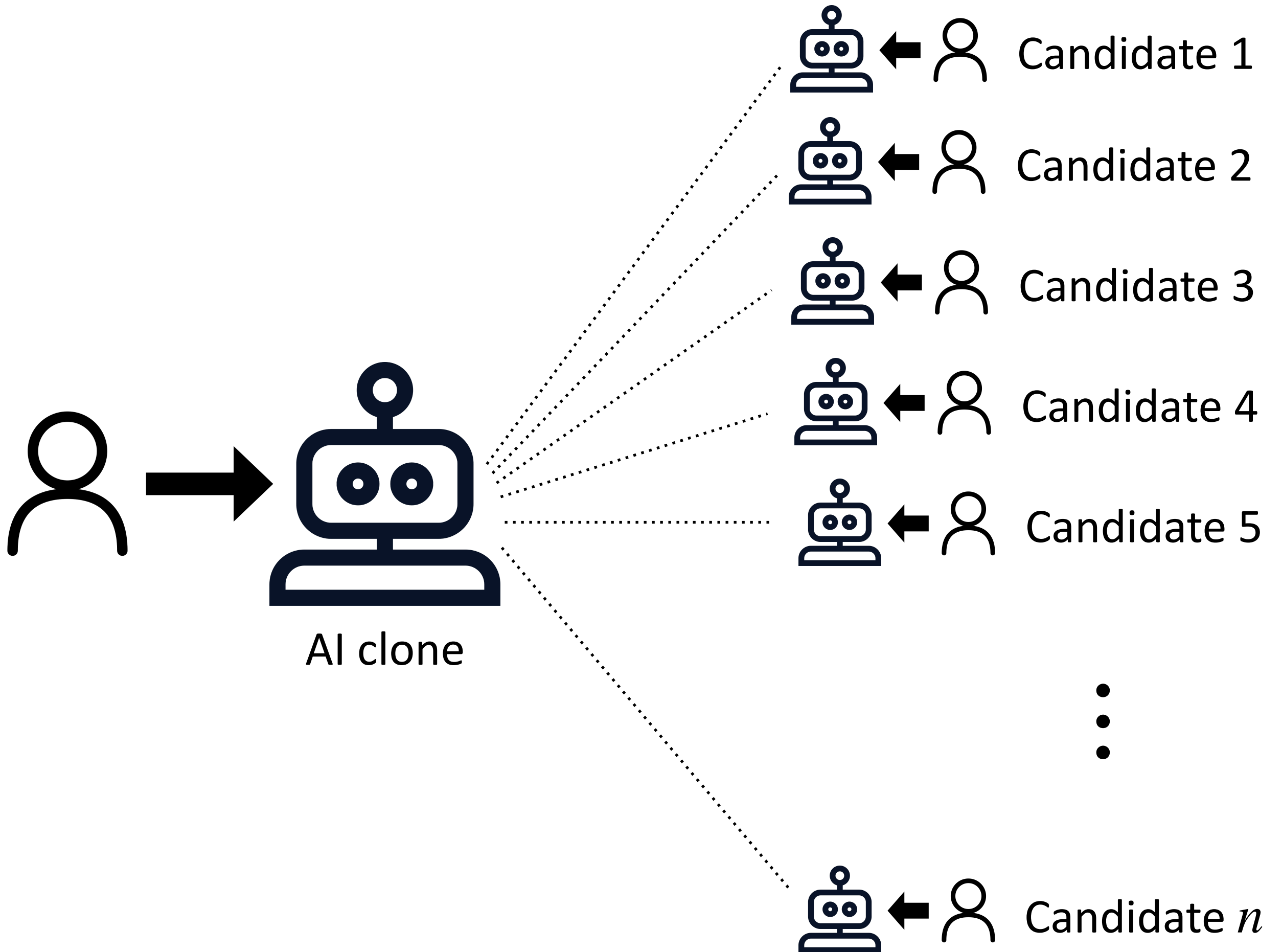
Are A.I. Clones the Future of Dating? I Tried Them for Myself.

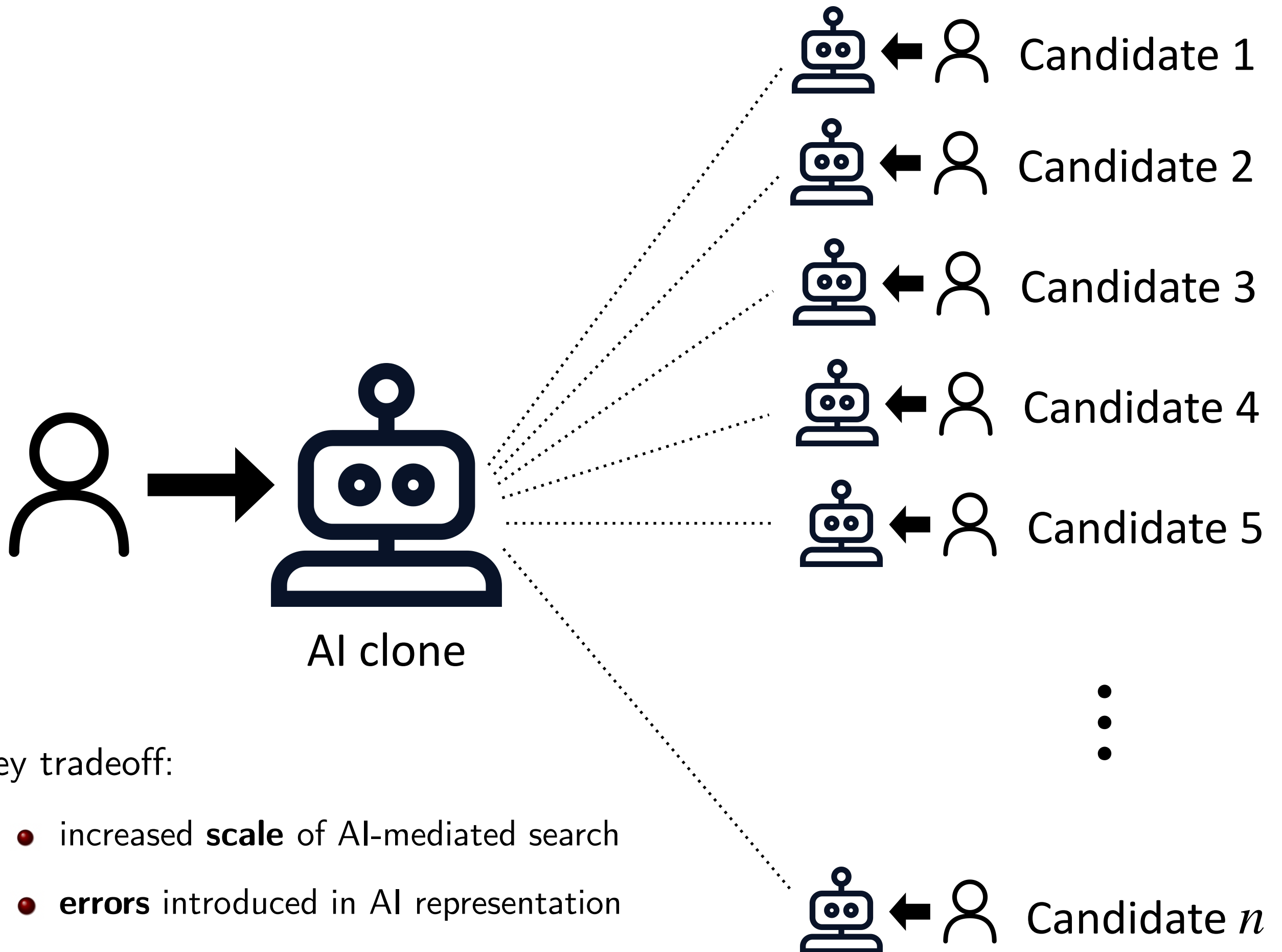
executives in the dating app industry have begun pitching a future in which people can create A.I. clones of themselves that date other clones and relay the results back to their human counterparts.

Dating Apps Suck. A.I. Clones Are Making Them Even Weirder.

‘My clone is on its way to date all of San Francisco in one month.’







Model

Setup

Individuals are represented by vectors in $\mathbb{R}^k$

- each entry is an attribute of the person

possible attributes:

extraversion

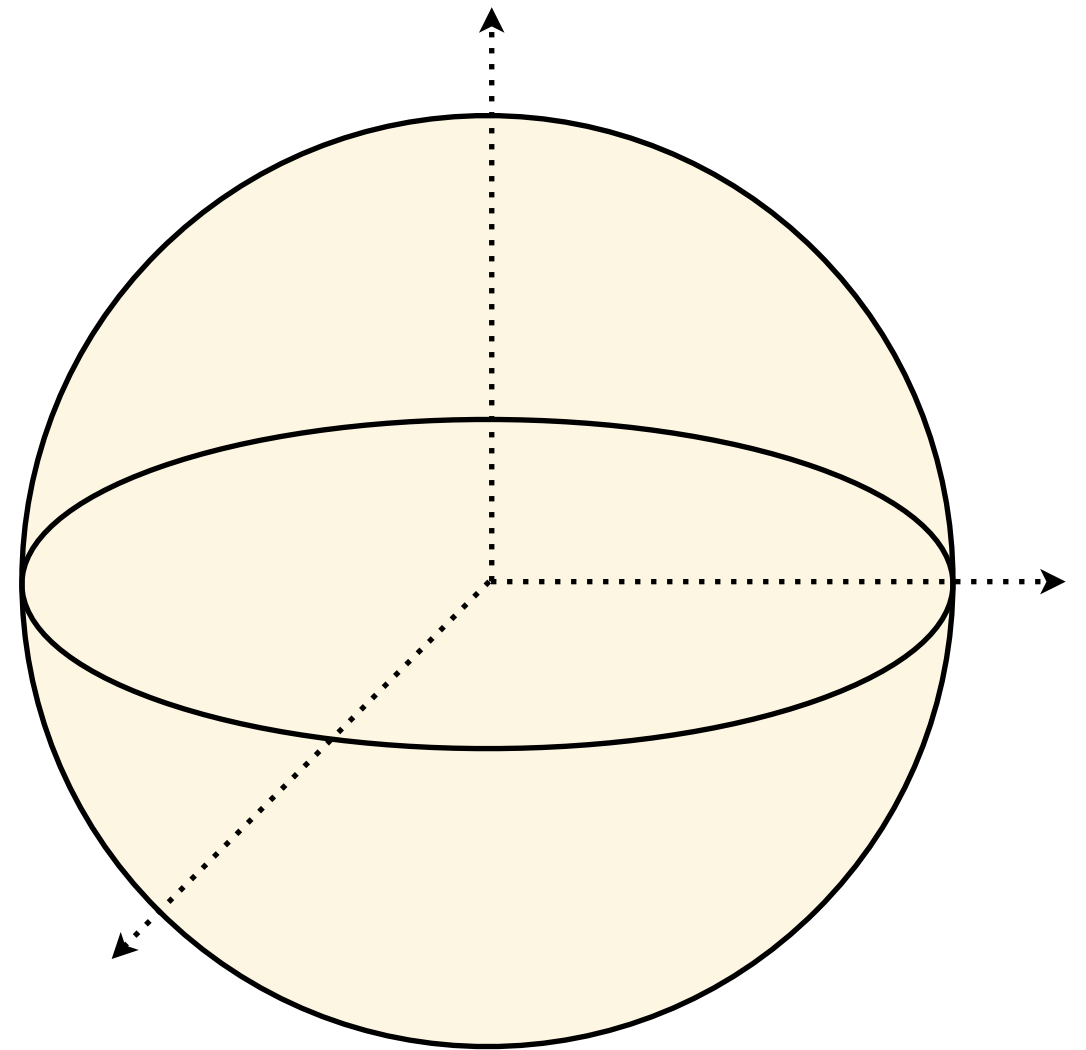
intellectual curiosity

relationship to time

style of humor

relationship to authority

speed of decision-making

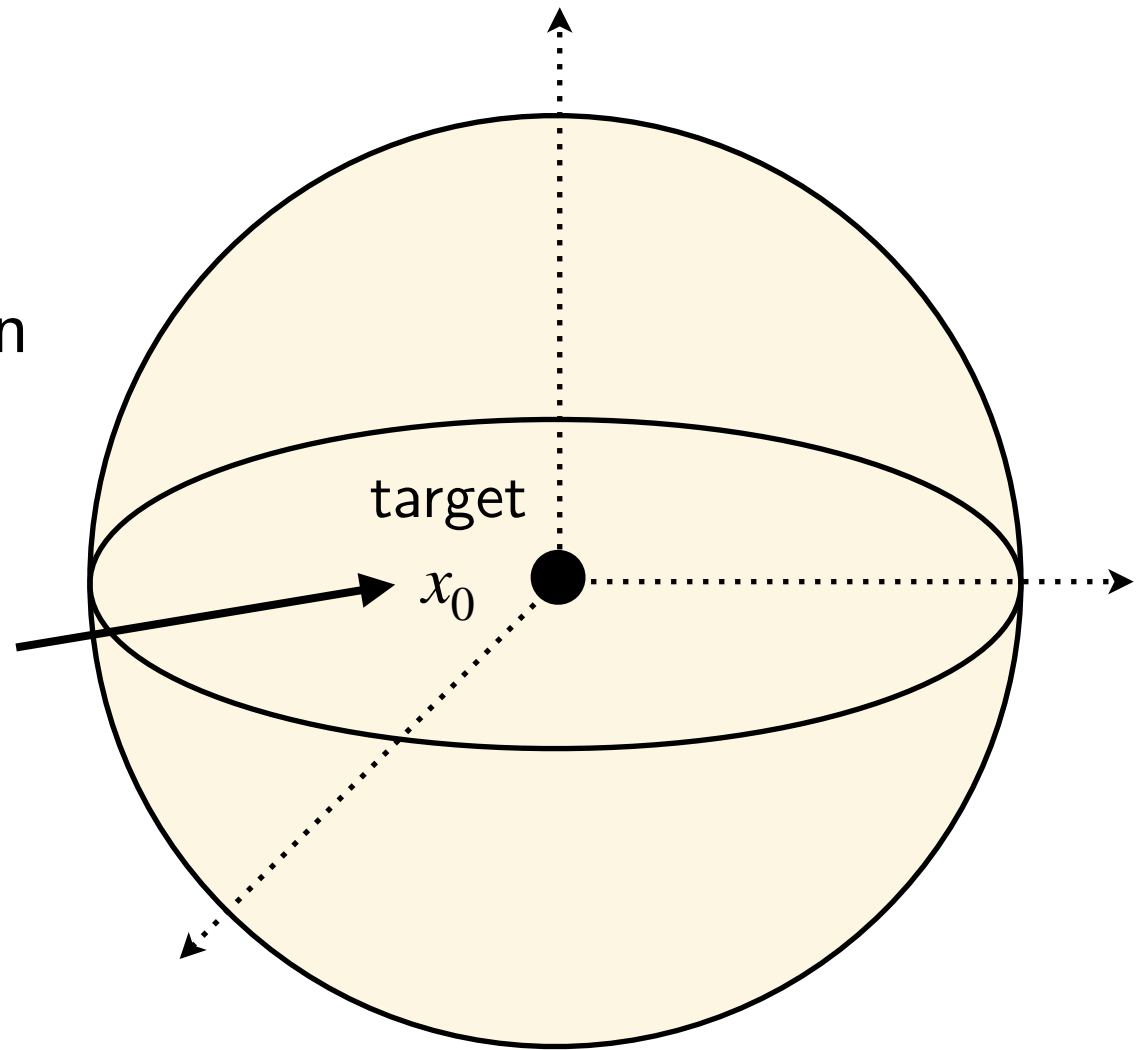


Setup

Individuals are represented by vectors in $\mathbb{R}^k$

- each entry is an attribute of the person

The searcher's ideal match is normalized to the origin and denoted x_0 (the **target**)



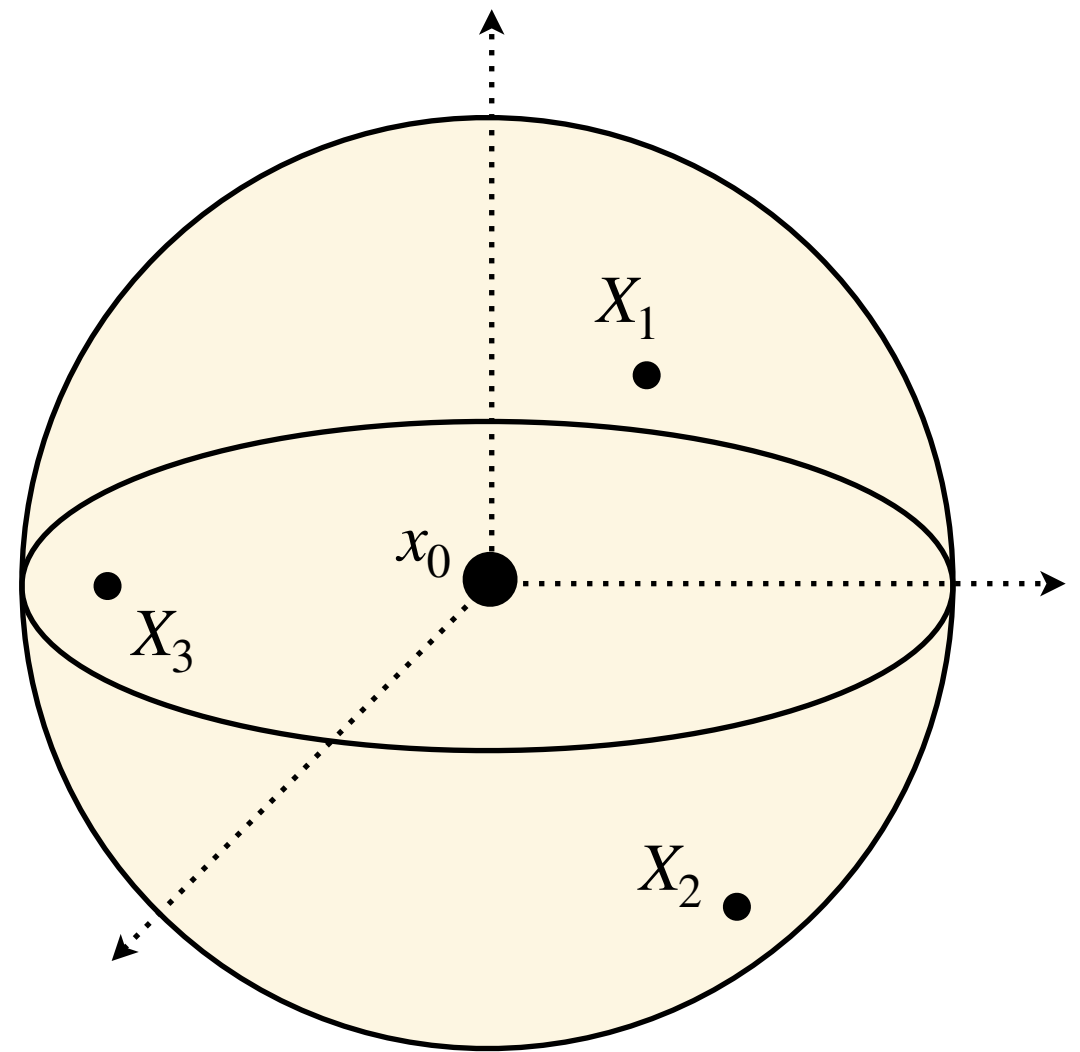
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People in the population $X_i \in \mathbb{R}^k$ are drawn uniformly at random from the unit ball B_k



Setup

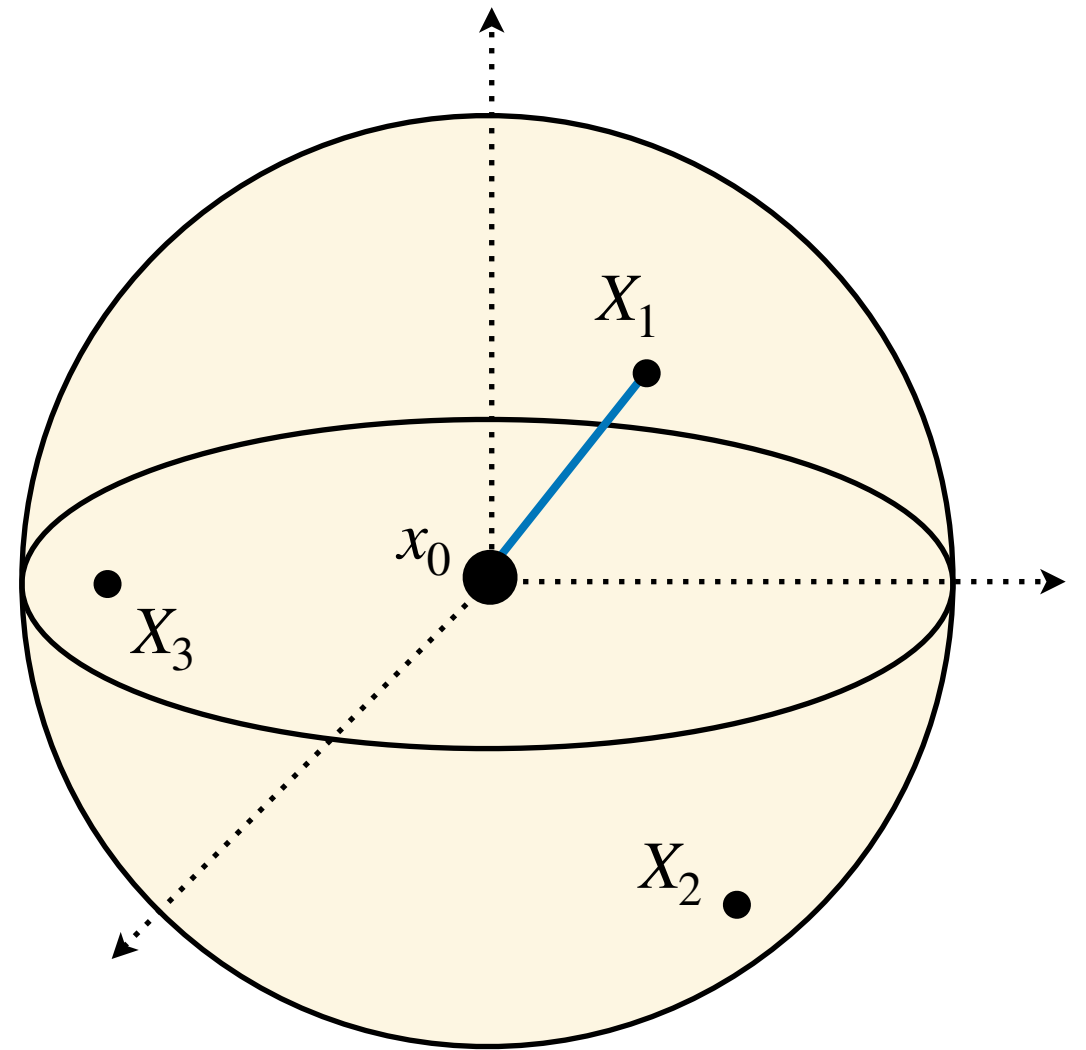
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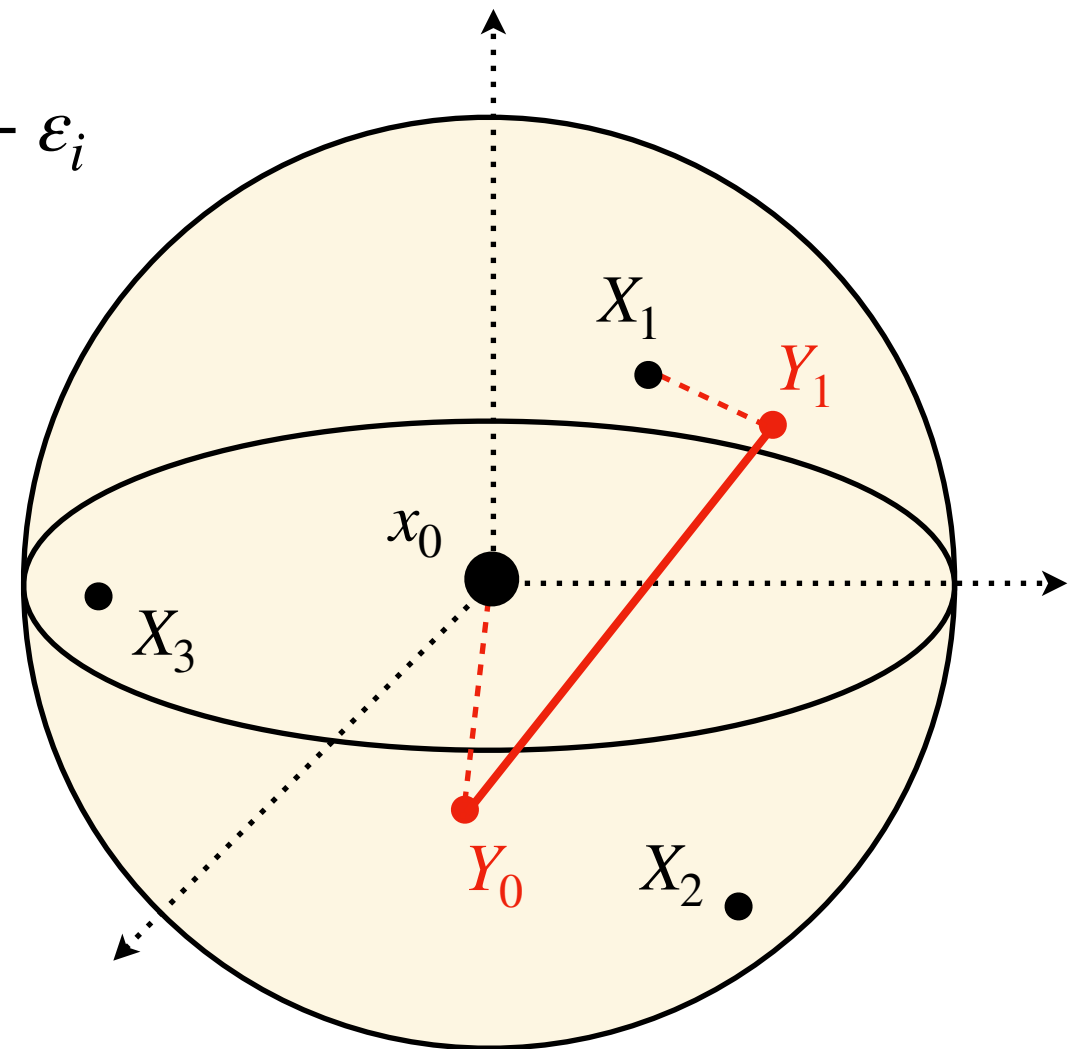
The searcher receives utility $-\|X_i\|$ from being matched to individual i



AI-Mediated Search

Under **AI-mediated** search,

- The searcher's target is represented by $Y_0 = x_0 + \varepsilon_0$
- Each individual i is represented by $Y_i = X_i + \varepsilon_i$
- The noise terms $\varepsilon_0, \varepsilon_i \sim \mathcal{N}(0, \sigma^2 I_k)$ are independent of one another
- Henceforth refer to Y_i and Y_0 as *AI representations*

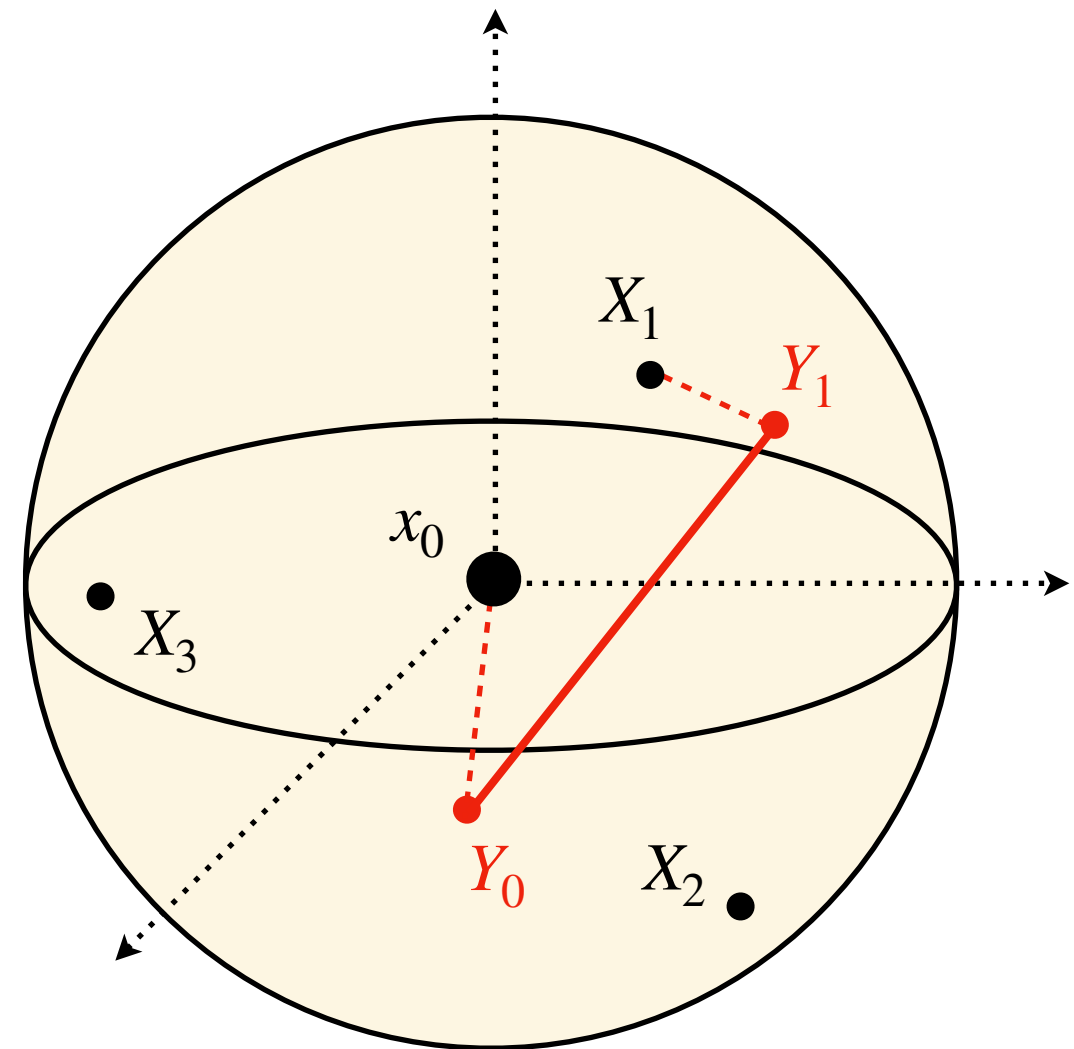


AI-Mediated Search

- The AI platform searches over n possible matches for the searcher
- The individual with the most compatible AI representation is

$$i_n^* = \operatorname{argmin}_{1 \leq i \leq n} \underbrace{\|Y_i - Y_0\|}_{\text{distance between the representation of the ideal point and the representation of individual } i}$$

distance between the representation
of the ideal point and the
representation of individual i



AI-Mediated Search

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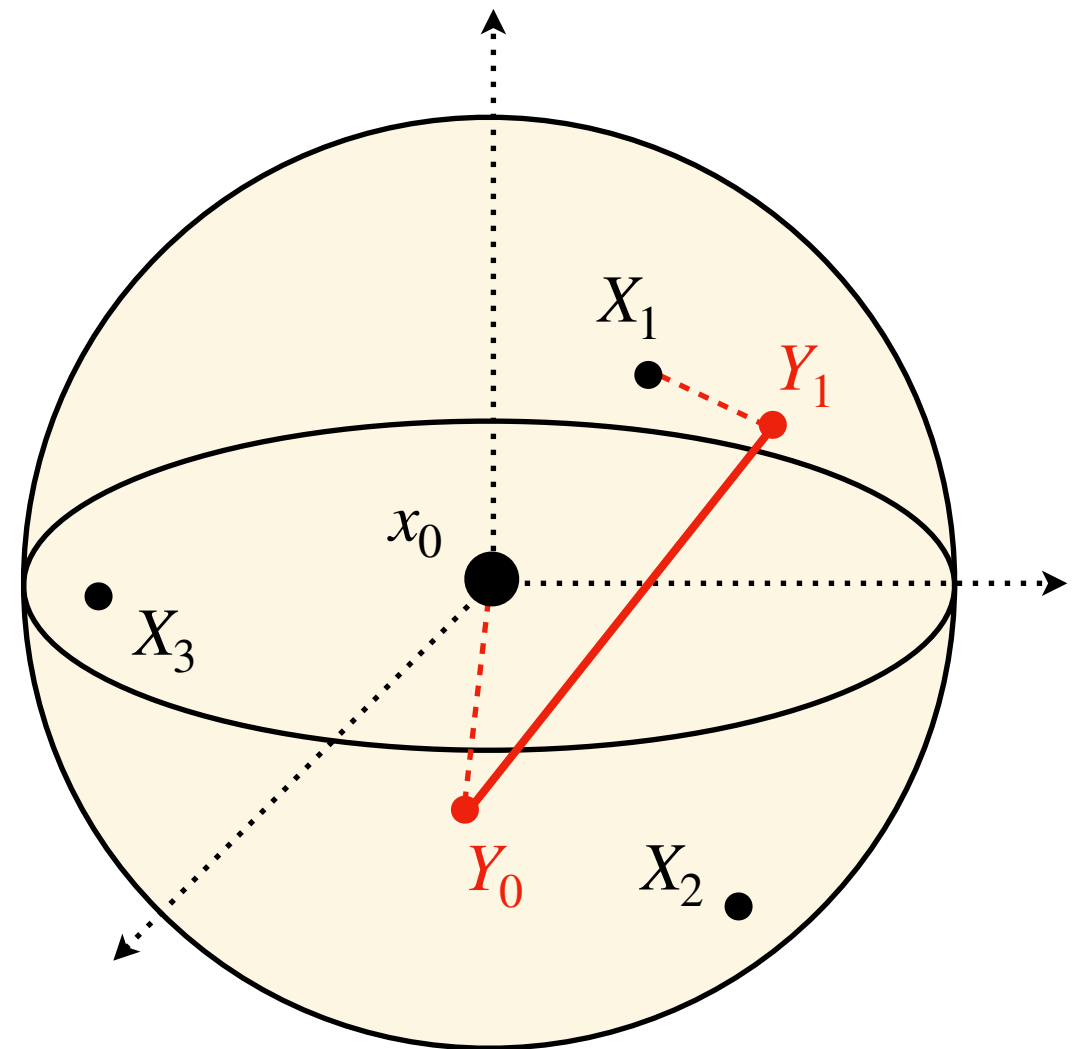
- The individual with the most compatible AI representation is

$$i_n^* = \operatorname{argmin}_{1 \leq i \leq n} \|Y_i - Y_0\|$$

- The searcher's expected utility is

$$u_k^{AI}(n) \equiv -E \left[\|X_{i_n^*}\| \right]$$

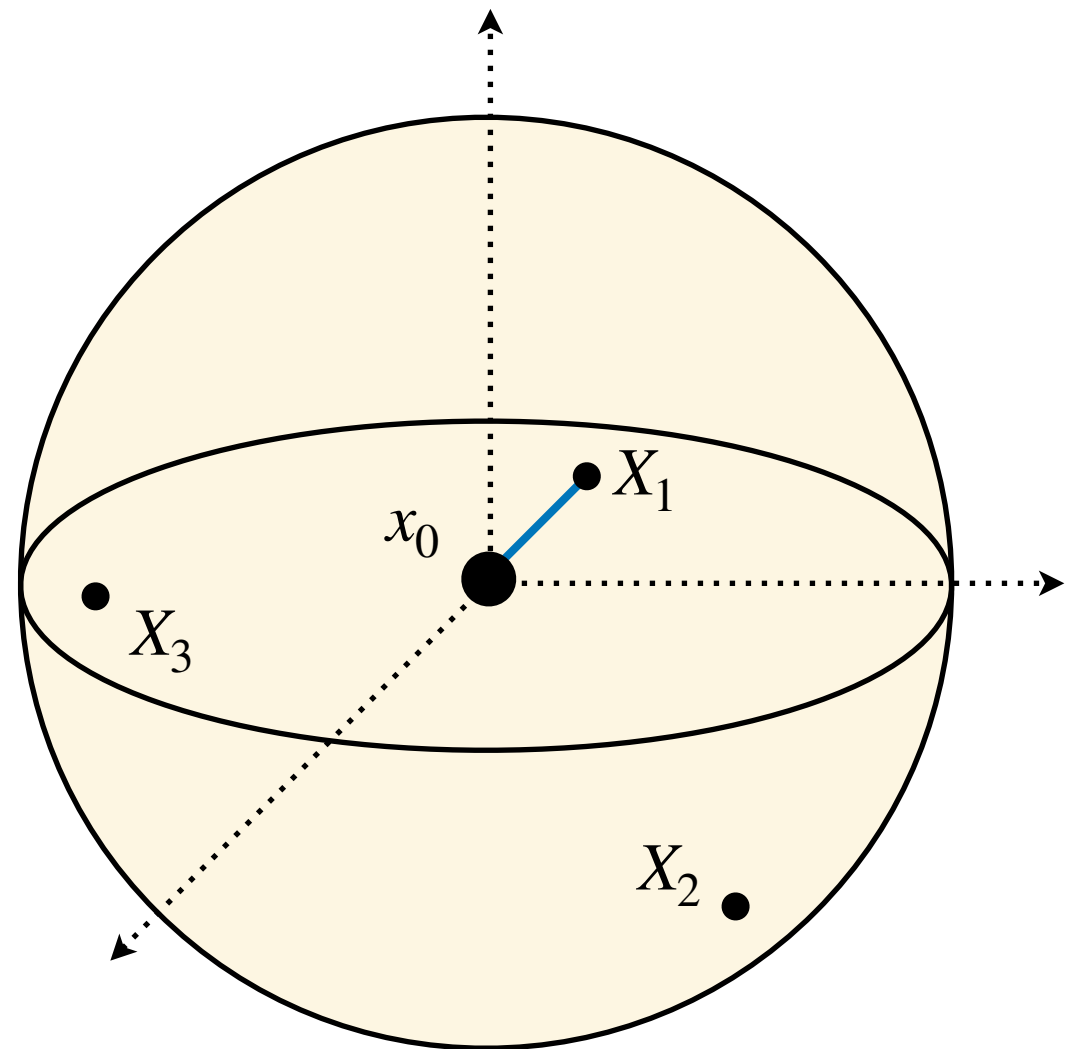
The selected match minimizes the distance between representations, but the searcher's payoff is based on the **true** distance between the underlying points



Benchmark: Best among m

- Suppose the searcher were to meet m individuals $X_1, \dots, X_m$ and match to the truly most compatible among them
- The searcher's expected utility from matching to this "best among m " is

$$u_k^B(m) \equiv - \mathbb{E} \left[\min_{1 \leq i \leq m} \|X_i\| \right]$$



AI-Equivalent Sample Size

Measure of the value of AI-mediated search:

Definition: For any number of dimensions k , the *AI-equivalent sample size* m_k^* is the smallest $m \in \mathbb{Z}_+$ satisfying

$$u_k^B(m) > u_k^{AI}(n) \quad \forall n \in \mathbb{Z}_+$$

if the inequality above is not satisfied for any finite m , then $m_k^* = \infty$.

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expected utility of best match
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expected utility of best match
on platform with n candidates
(AI-mediated search)

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$$\underbrace{u_k^B(m) > u_k^{AI}(n)}_{\text{inequality}} \quad \forall n \in \mathbb{Z}_+$$

if the inequality above is not satisfied for any finite m , then $m_k^* = \infty$.

matching to the best candidate among m draws beats the AI platform, no matter how large it is

Results

Main Results

Proposition: For every number of dimensions $k \in \mathbb{Z}_+$, the AI-equivalent sample size m_k^* is finite

- The value to increasing scale is capped

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Theorem: For all k sufficiently large, the AI-equivalent sample size is $m_k^* = 2$

- When there are sufficiently many relevant attributes, being on the platform is no better than meeting two people and matching to the better among them
- Underscores the limits of AI representations for complex search problems

Proof Idea/Intuition

Expected Utility:

Benchmark $u_k^B(m) \equiv - \mathbb{E} \left[\min_{1 \leq i \leq m} \|X_i\| \right]$

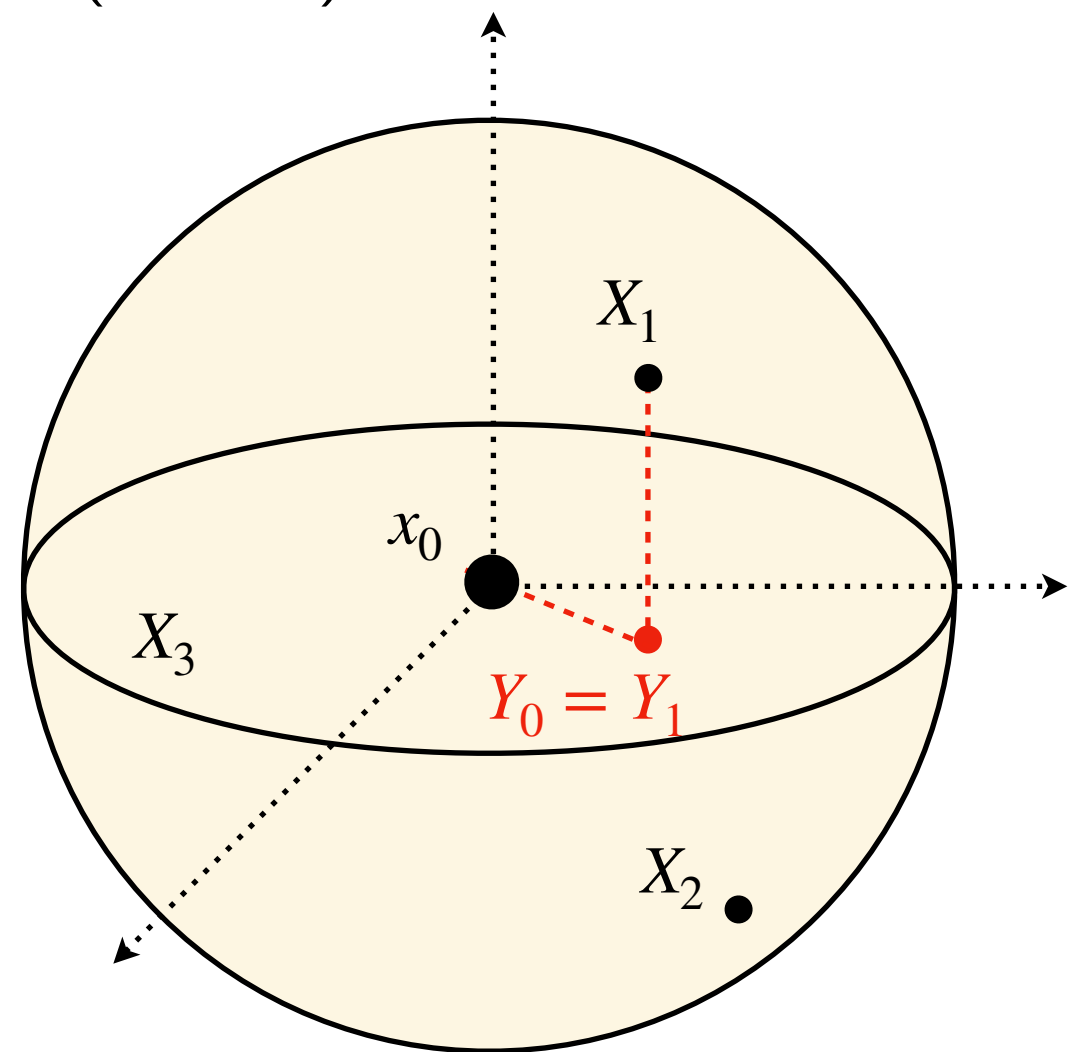
AI-mediated search $u_k^{AI}(n) \equiv - \mathbb{E} \left[\|X_{i_n^*}\| \right]$

need to show: $u_k^B(2) > u_k^{AI}(n)$ for all n

Proof Idea/Intuition

Define $u_k^{AI}(\infty) \equiv -E(\|X_i\| : \|Y_i - Y_0\| = 0)$

- The expected distance to an individual whose AI representation perfectly matches the target representation
- Can interpret as completely saturated limit ($n = \infty$)



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- This follows because the pair $(\|X_i\|, \|Y_i - Y_0\|)$ has the monotone likelihood ratio property (MLRP)

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Next argue that $u_k^{AI}(\infty)$ converges to $u_k^B(1)$ as k grows large

- Getting your best match on the infinite AI platform is eventually as bad as matching to the first person you meet
- So matching to the best among **two** is better

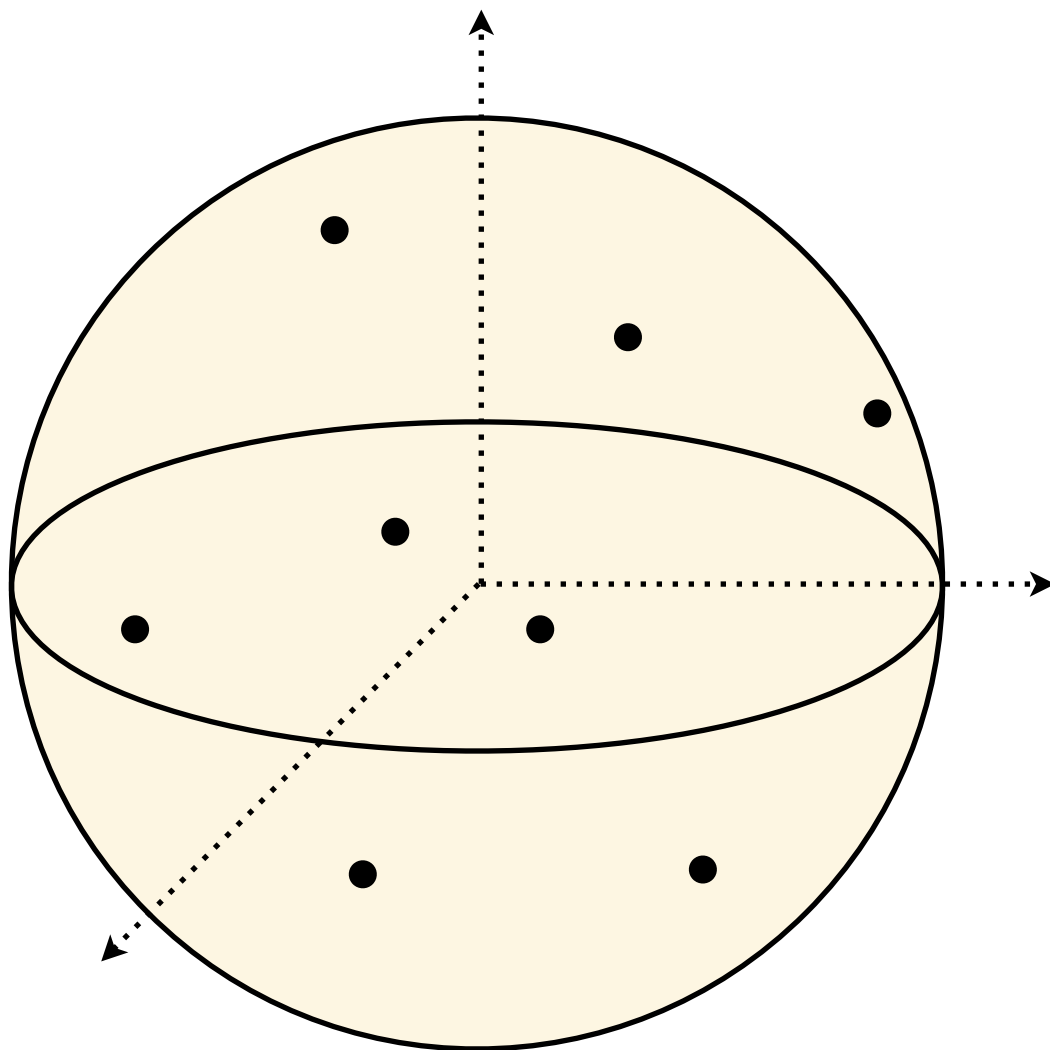
Proof Idea/Intuition

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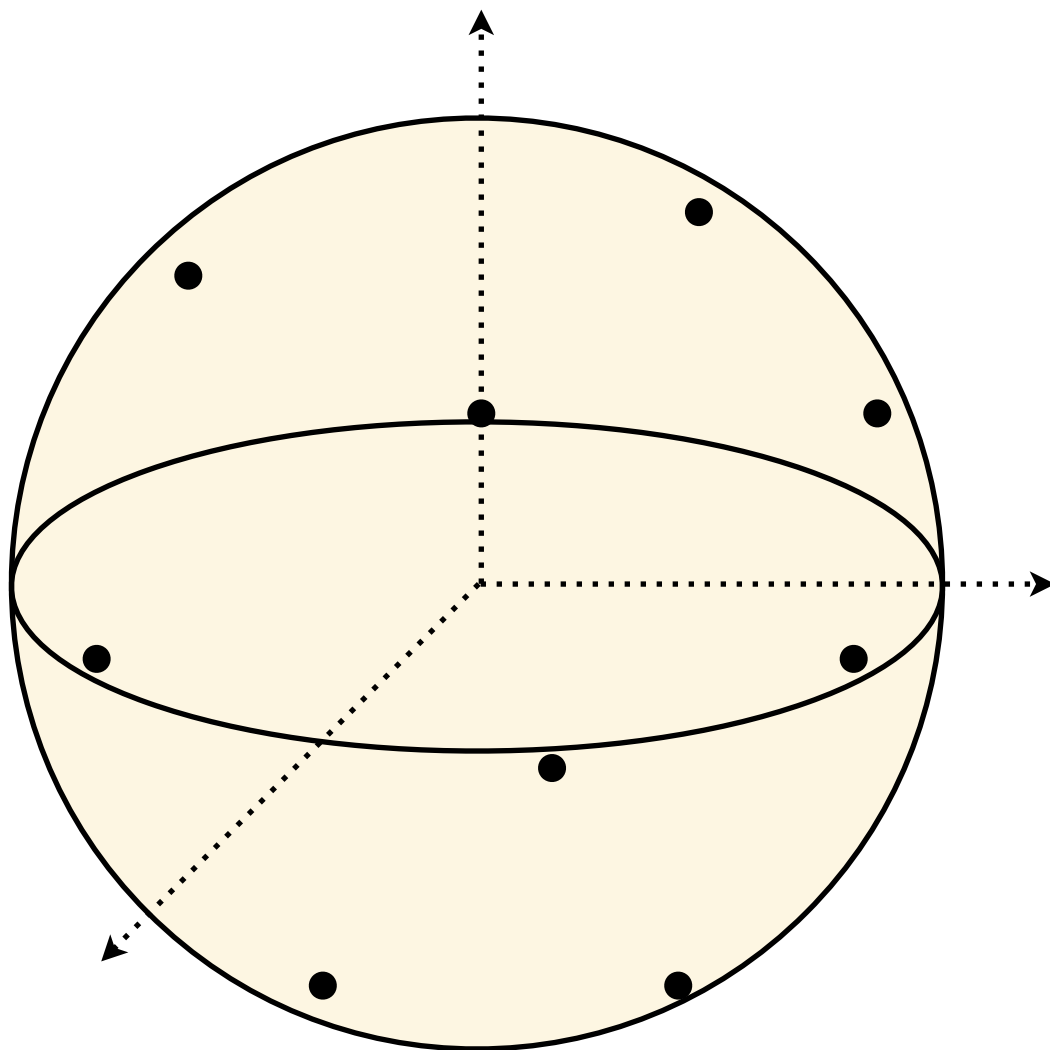
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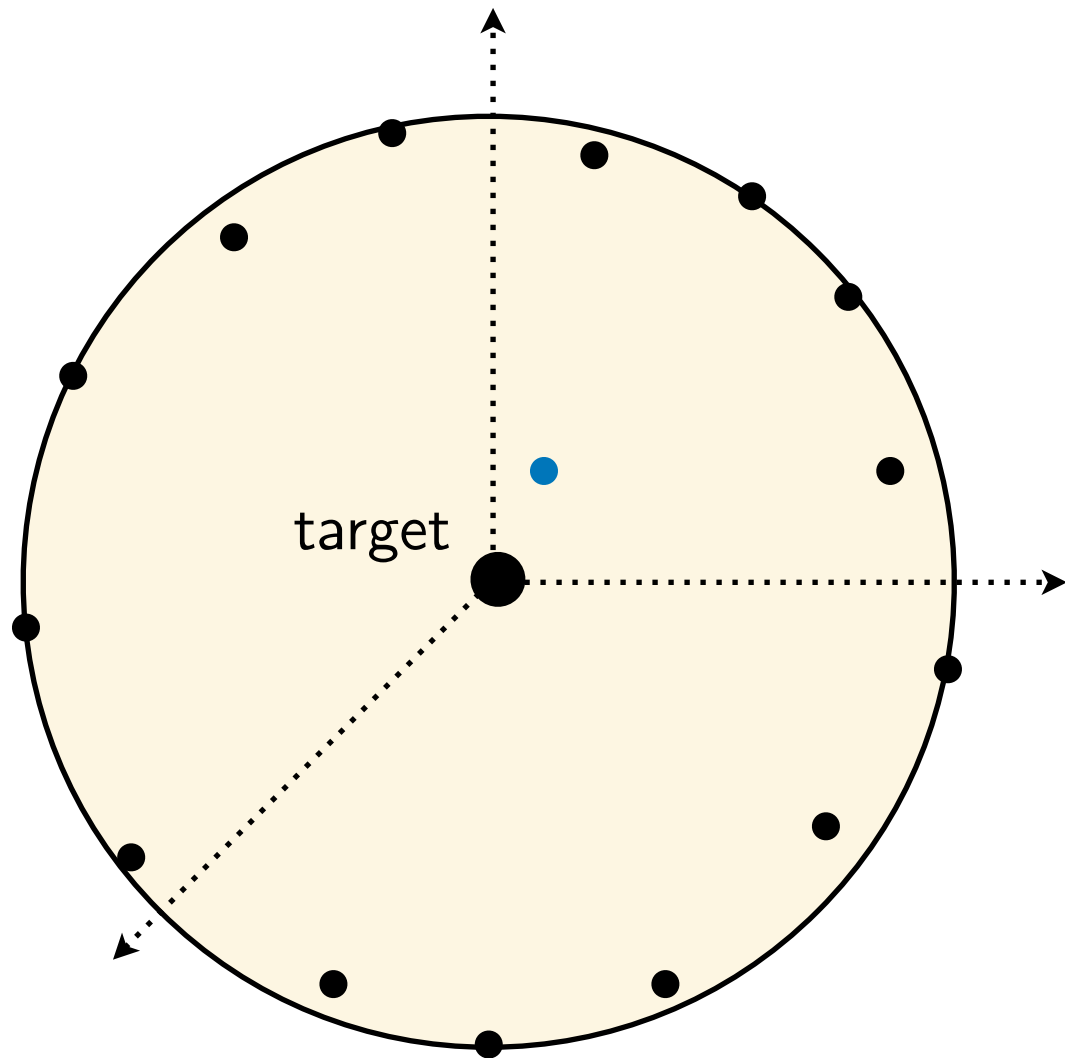
In fact, in high dimensions:

- points amass on the boundary of the unit ball
- distances between them concentrate around their expectations

i.e., people are increasingly unique and increasingly unlikely to exactly match the target

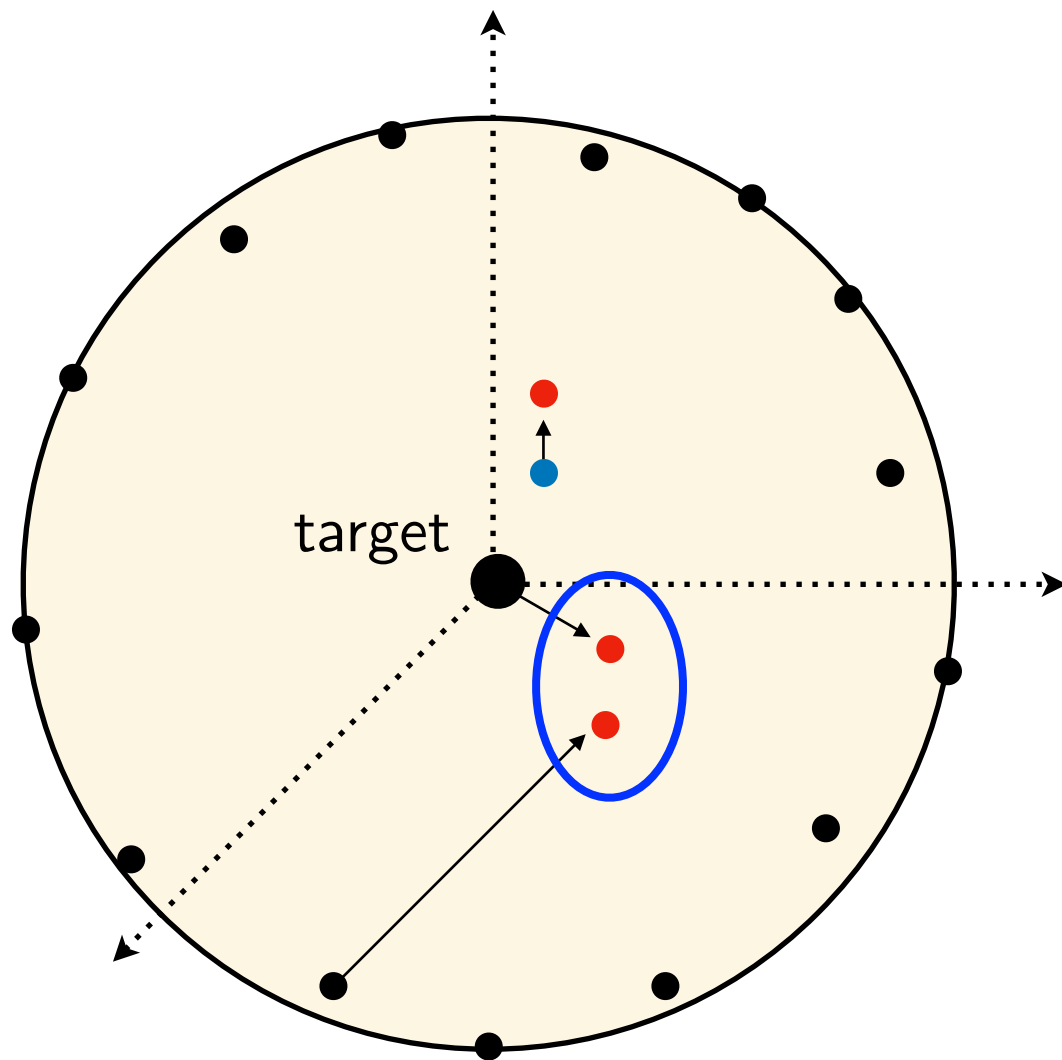
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Proof Idea/Intuition

Why does $u_k^{AI}(\infty)$ converge to $u_k^B(1)$?



- Best match increasingly driven by realization of the noise terms
- Eventually as if just selecting one candidate at random (i.e., $d_k^B(1)$)

How Large is k ?

- In a simulation with $\sigma = 0.05$, the result holds for $k = 150$ dimensions

How Large is k ?

- In a simulation with $\sigma = 0.05$, the result holds for $k = 150$ dimensions
- How many dimensions of personality are relevant for identifying
 - good spousal matches?
 - good employees?
- This is an empirical question

How Large is k ?

- In a simulation with $\sigma = 0.05$, the result holds for $k = 150$ dimensions
- How many dimensions of personality are relevant for identifying
 - good spousal matches?
 - good employees?
- This is an empirical question
- The main takeaway is that **the complexity of match compatibility matters** for how we should think about human vs. AI search
- The value of AI representations can differ across search problems depending on what the relevant k is

Creative Ownership in the Age of AI

Annie Liang
(Northwestern)

Jay Lu
(UCLA)

Introduction

Have already discussed how generative AI systems can now create images, novels, persons, etc.

These capabilities depend crucially on having been trained on a large body of human-authored work.

AI creations can closely imitate the style of those original works without copying content, a capability now central to ongoing litigation.

- Andersen v. Stability AI
 - Disney Enterprises v. Midjourney
 - Universal Music Group Recordings v. Suno
- + many others

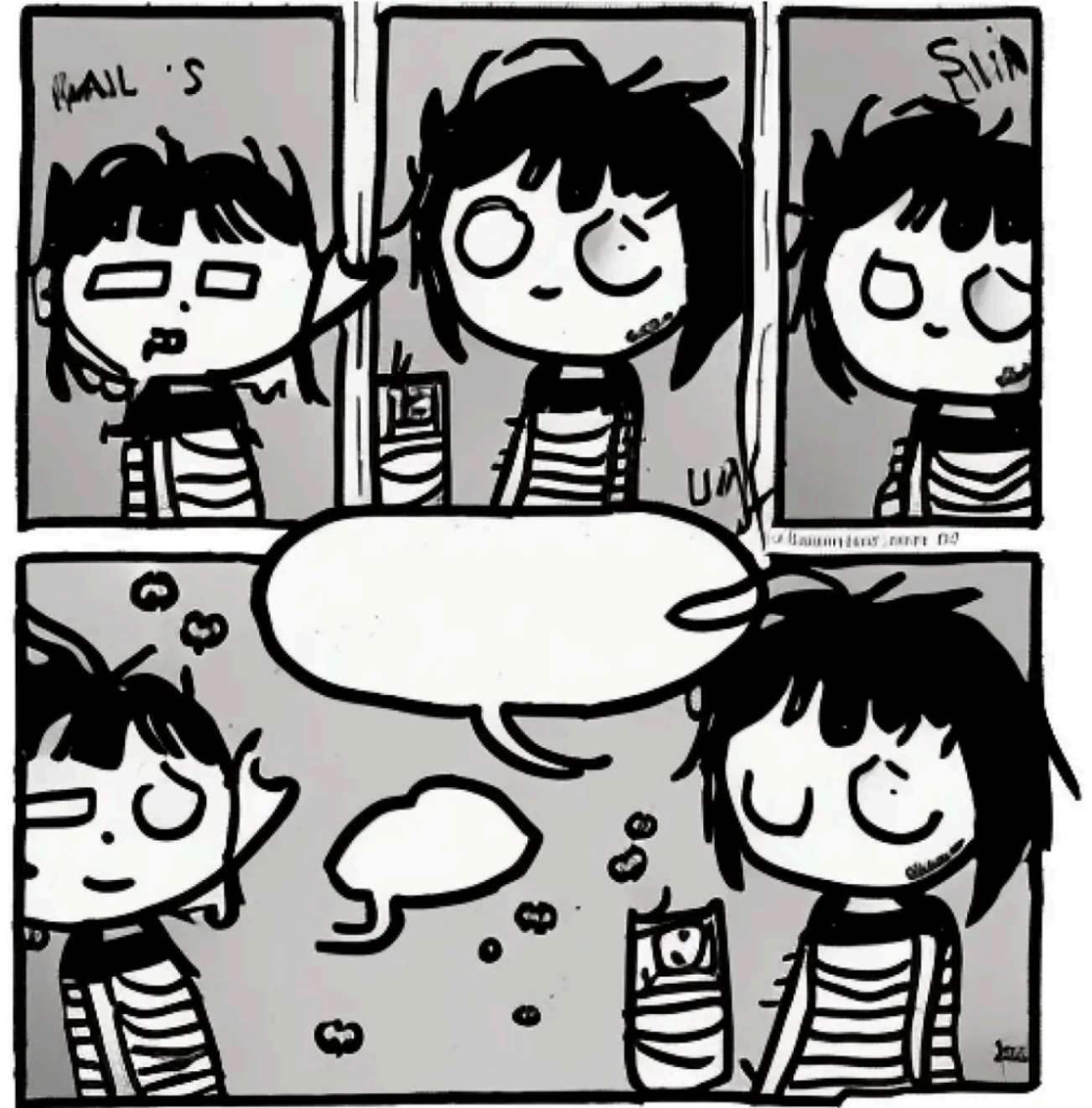
Introduction

A recent example of my work



Sarah Andersen, *NYTimes*, 2022

An A.I. generated image that I created when I used my name as a prompt



It's not perfect — but it has captured the signature elements of my drawing style. The black bangs, striped shirt and wide eyes are immediately recognizable. As A.I. gets increasingly sophisticated, I am growing more concerned about what's possible.

Legal Context

Existing copyright doctrine applies to GenAI at two stages:

1. **Training** on copyrighted works may violate copyright law.

- Possibly too strong: human authors learn by reading copyrighted books.
- May overly constrain AI capabilities to not allow it to learn in a similar way
- Moreover, fair use doctrine suggests that training on copyrighted works may be OK if the use is sufficiently “transformative”

2. **Generated outputs** may violate copyright law if they are “substantially similar” to an existing work.

- Copyright law has historically drawn a line between protectable **expression** and unprotectable **style** to preserve creative competition and follow-on innovation.
- But did not account for the emergence of AI systems that could easily, flexibly, quickly mimic style

This Paper

Existing notions of copyright and intellectual property **may not be appropriate for a post-generative AI world.**

We propose a new criterion for evaluating when an output is derivative of an existing work:

Would the output have been generable without that existing work in its training corpus?

Model

Setup

- A *creation* is a vector $x \in \mathbb{R}^d$
- A *corpus* is any (Borel measurable) subset of creations $C \subset \mathbb{R}^d$
- Let $\mathcal{C}$ denote the set of all possible corpora
- A *generator* is a technology for producing new creations from an existing corpus.

Generators

Definition: The map $g : \mathcal{C} \rightarrow \mathcal{C}$ is a *generator* if for all $C, C' \in \mathcal{C}$ it satisfies:

1. Preservation: $C \subseteq g(C)$ — *reproduces all input creations*
2. Monotonicity: $C \subseteq C'$ implies $g(C) \subseteq g(C')$ — *weakly expands the set of generable creations when the original corpus expands*
3. Idempotence: $g(g(C)) = g(C)$ — *running it on its output produces no new creations*

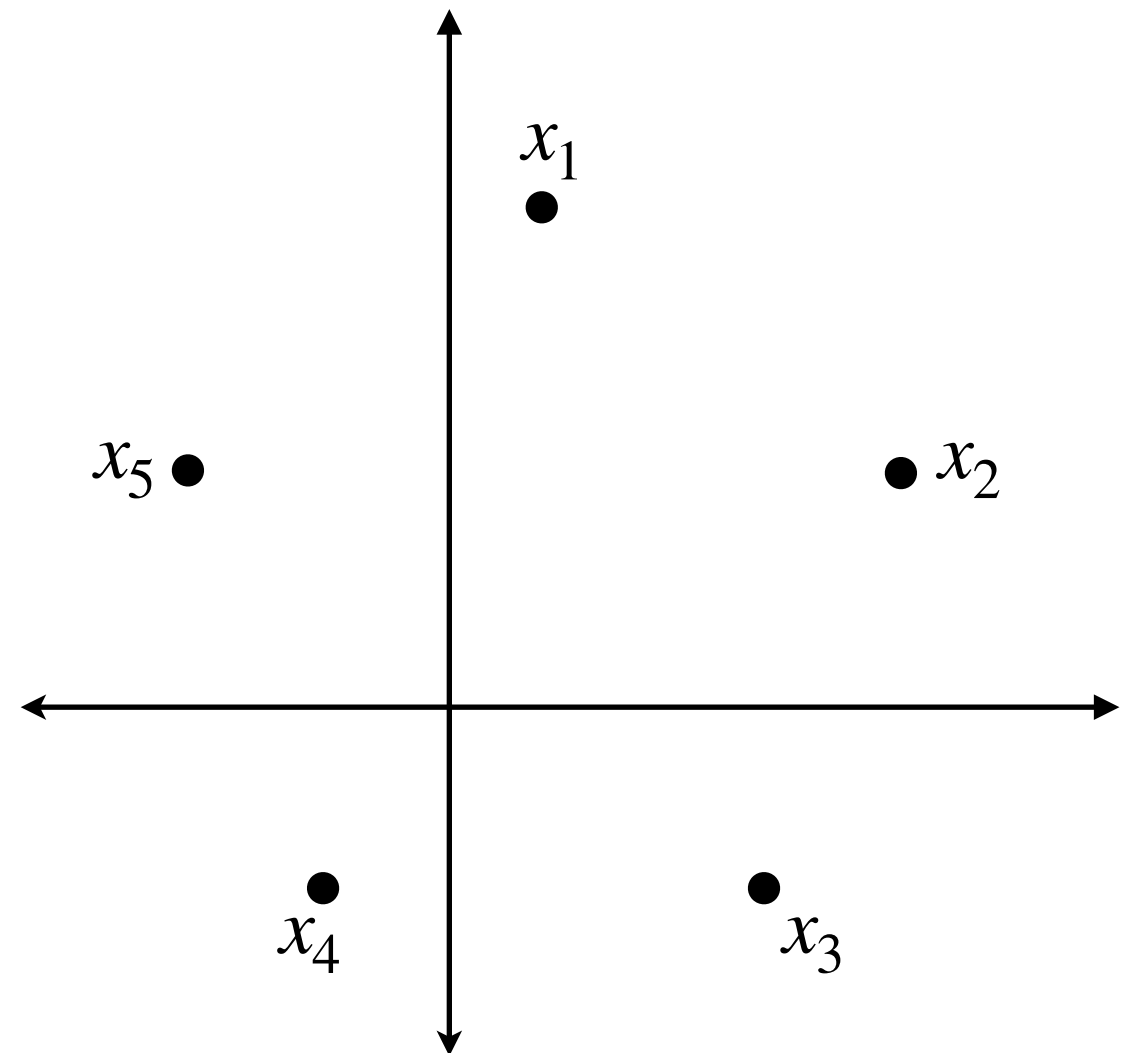
These are also known as closure operators.

We interpret $g(C)$ interpreted as the set of all possible creations given the initial training corpus C .

Example Generators

Convex Hull Generator: $g_{conv}(C) = \text{Conv}(C)$

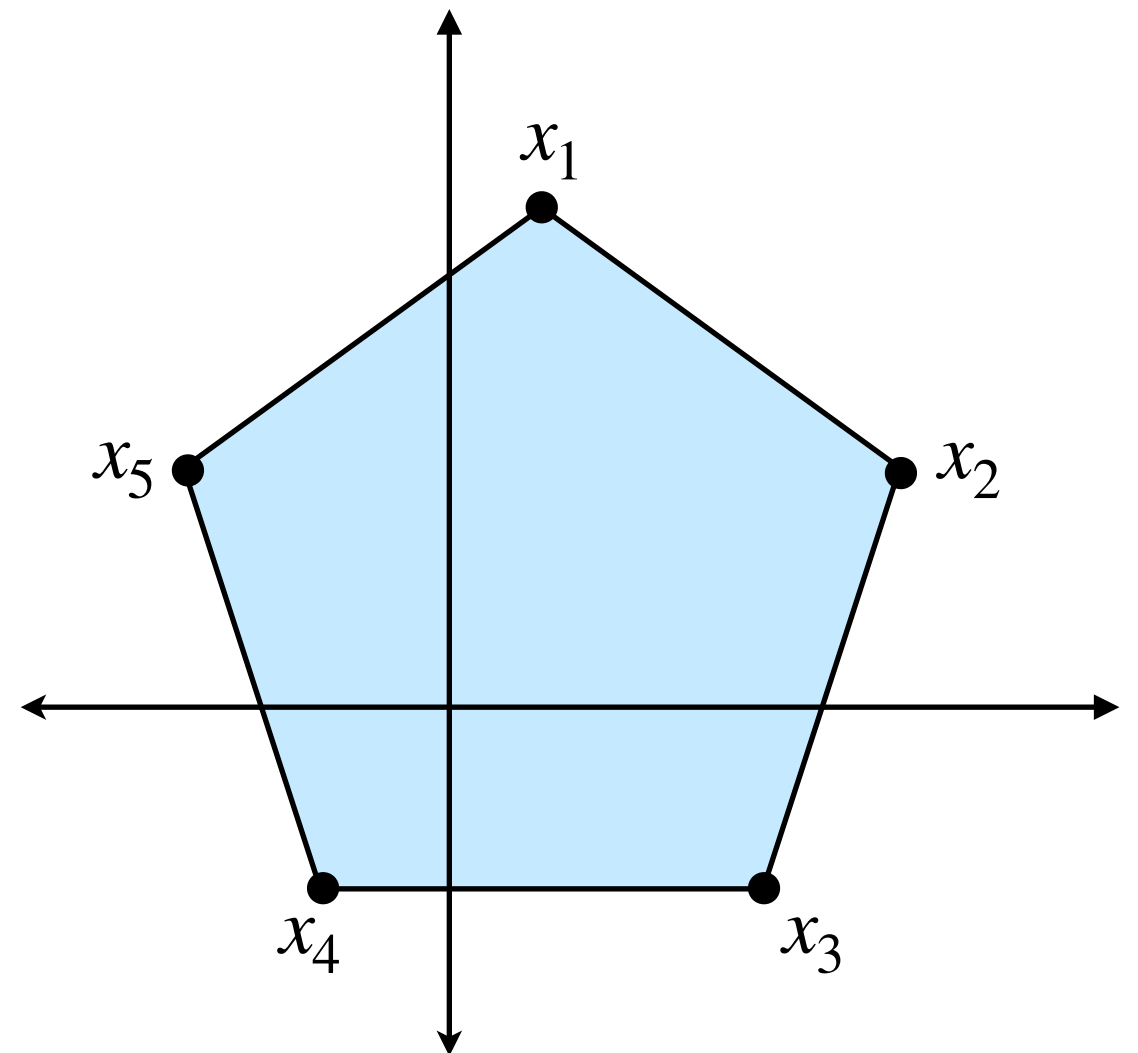
- Generate any convex combination of existing creations



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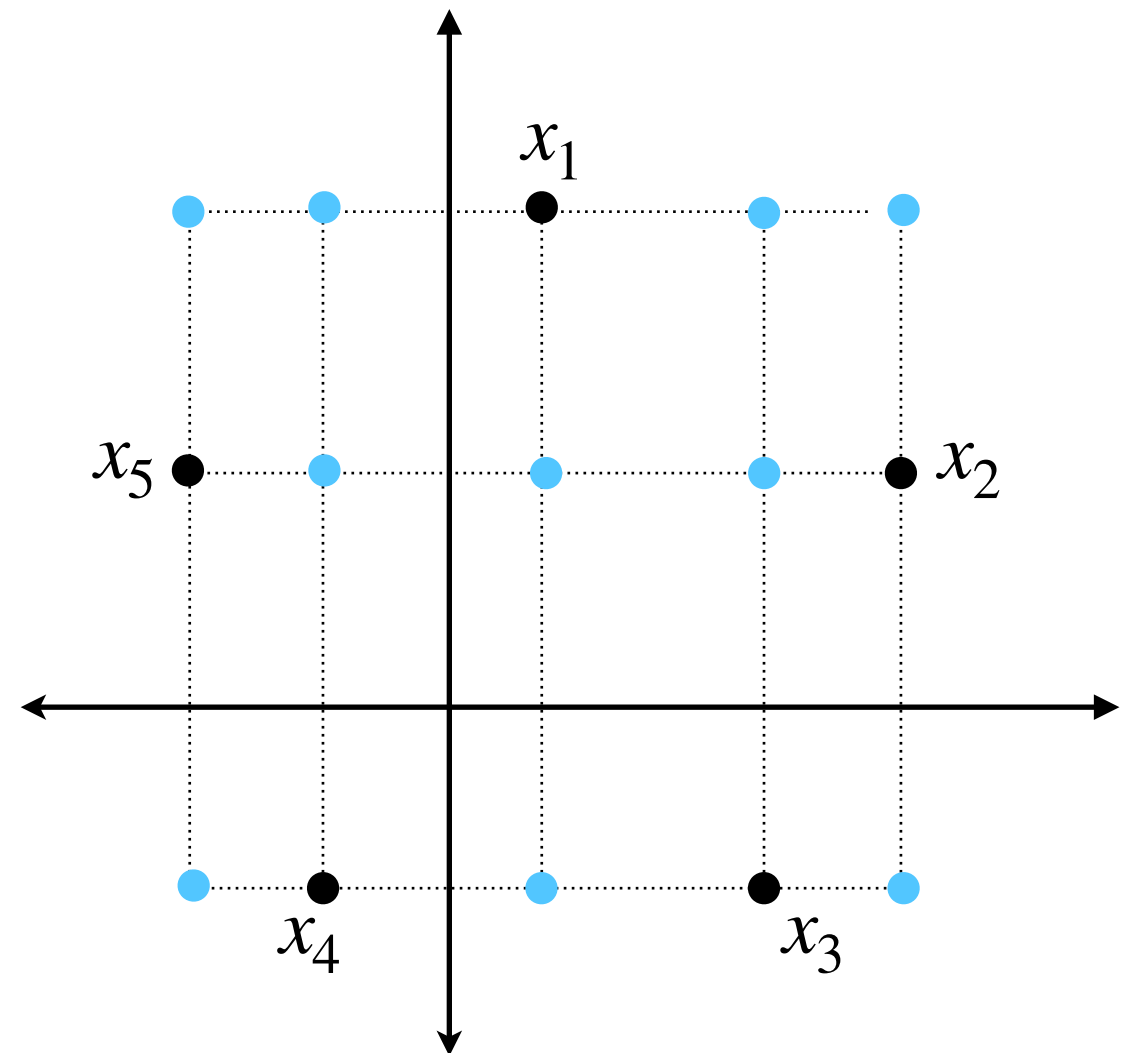
Example Generators

Convex Hull Generator: $g_{conv}(C) = \text{Conv}(C)$

- Generate any convex combination of existing creations

Splice Generator: $g_{splice}(C) := \{x \in \mathbb{R}^d : \forall k = 1, \dots, d, \text{ there is a } x' \in C \text{ s.t. } x_k = x'_k\}$.

- Recombines features of existing creations



Example Generators

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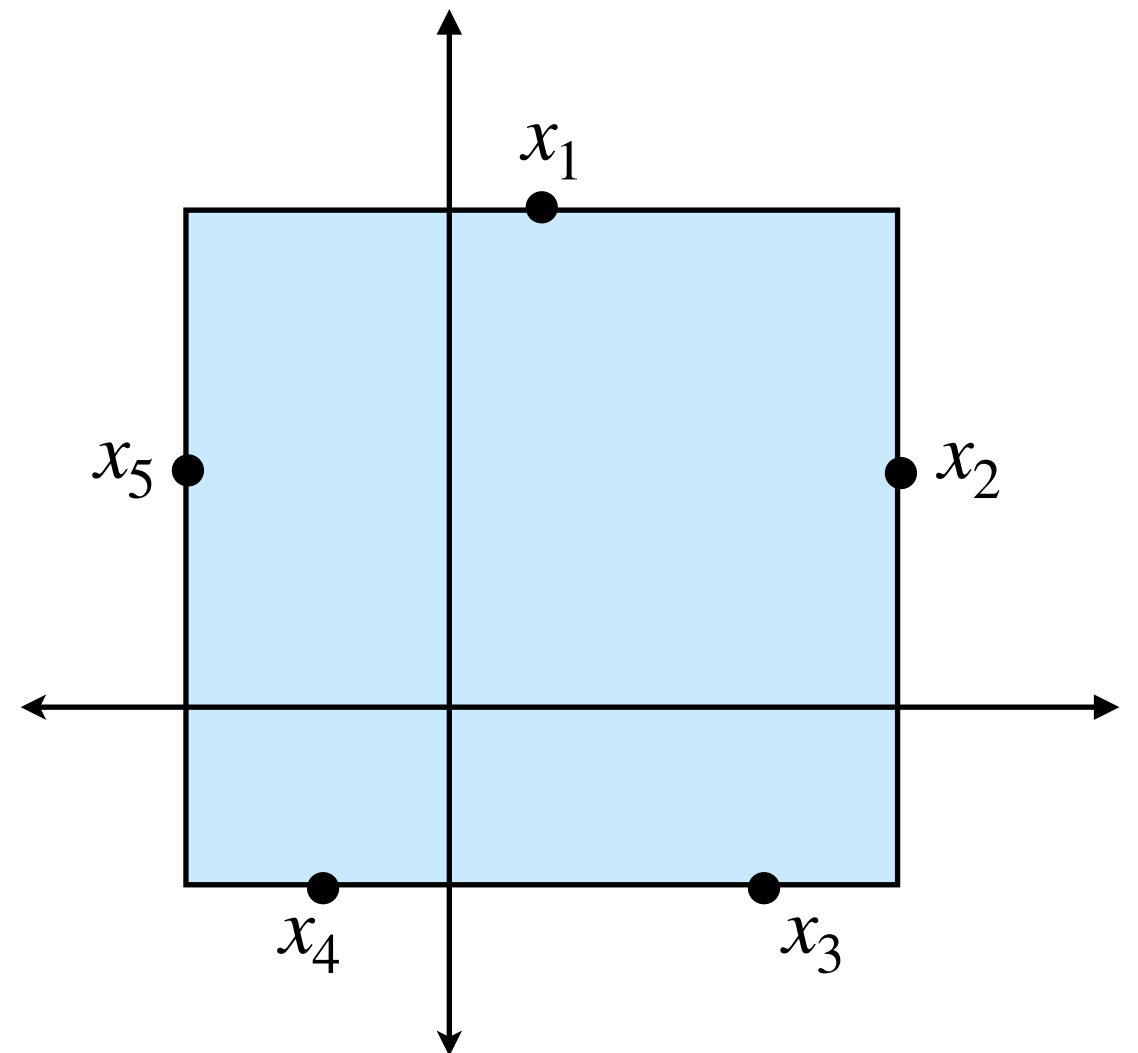
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- Recombines features of existing creations

Box Generator: $g_{box}(C) = g_{conv}(g_{splice}(C))$

- Composes the convex hull and splice generator



Convex-Valued Generators

Definition: A generator g is *convex-valued* if $g(C)$ is convex for every $C \in \mathcal{C}$.

- Includes the convex-hull generator and box generator (but not the splice generator)

Result: The following are equivalent:

1. g is convex-valued
2. $g = g_{conv} \circ g$
3. $g = g \circ g_{conv}$

- Thus can interpret restriction to convex-valued generators as an assumption that the convex-hull generator is available
- Some of our results will make this restriction

Permissible Generation

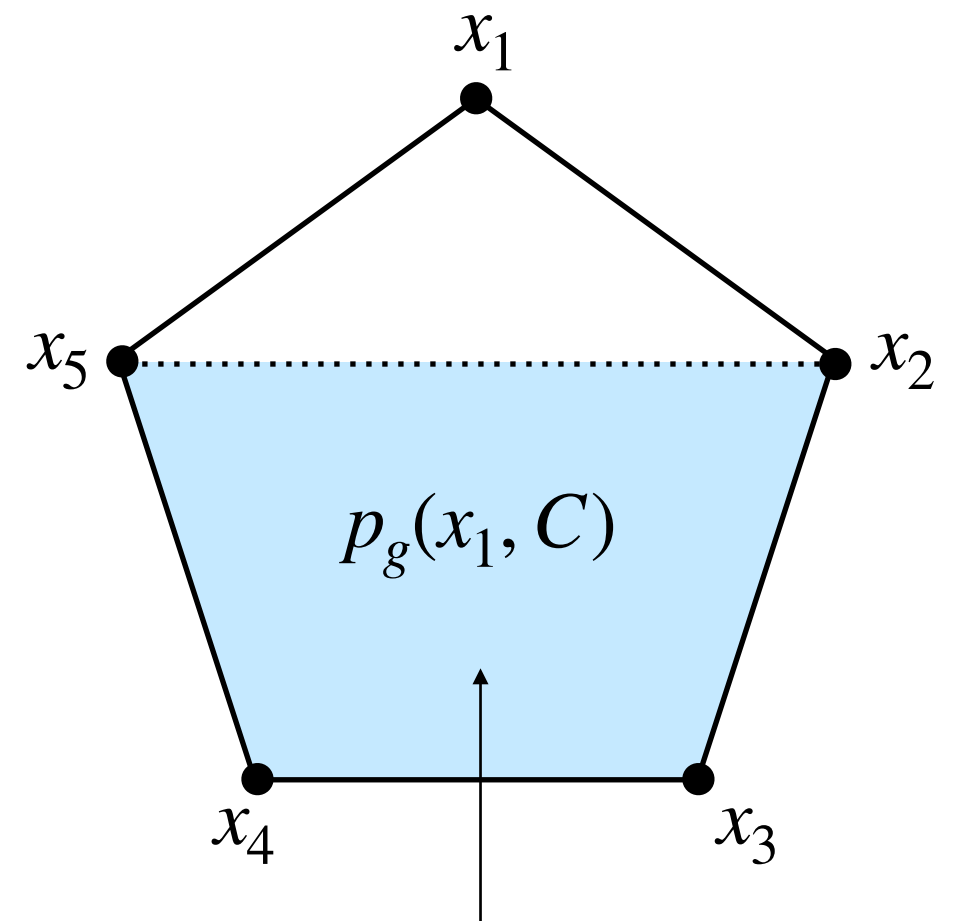
- Our proposed criterion specifies when a generation critically depends on the presence of some existing work.
- Would the creation still be generable without that existing work in the corpus?

Permissible Generation

Definition: Fix a generator g and corpus C . For each existing creation $x \in C$:

1. The x -**permissible set** is $p_g(x, C) := g(C \setminus \{x\})$, i.e., all creations that are generable from $C \setminus \{x\}$

Convex hull generator:



x_1 -permissible set

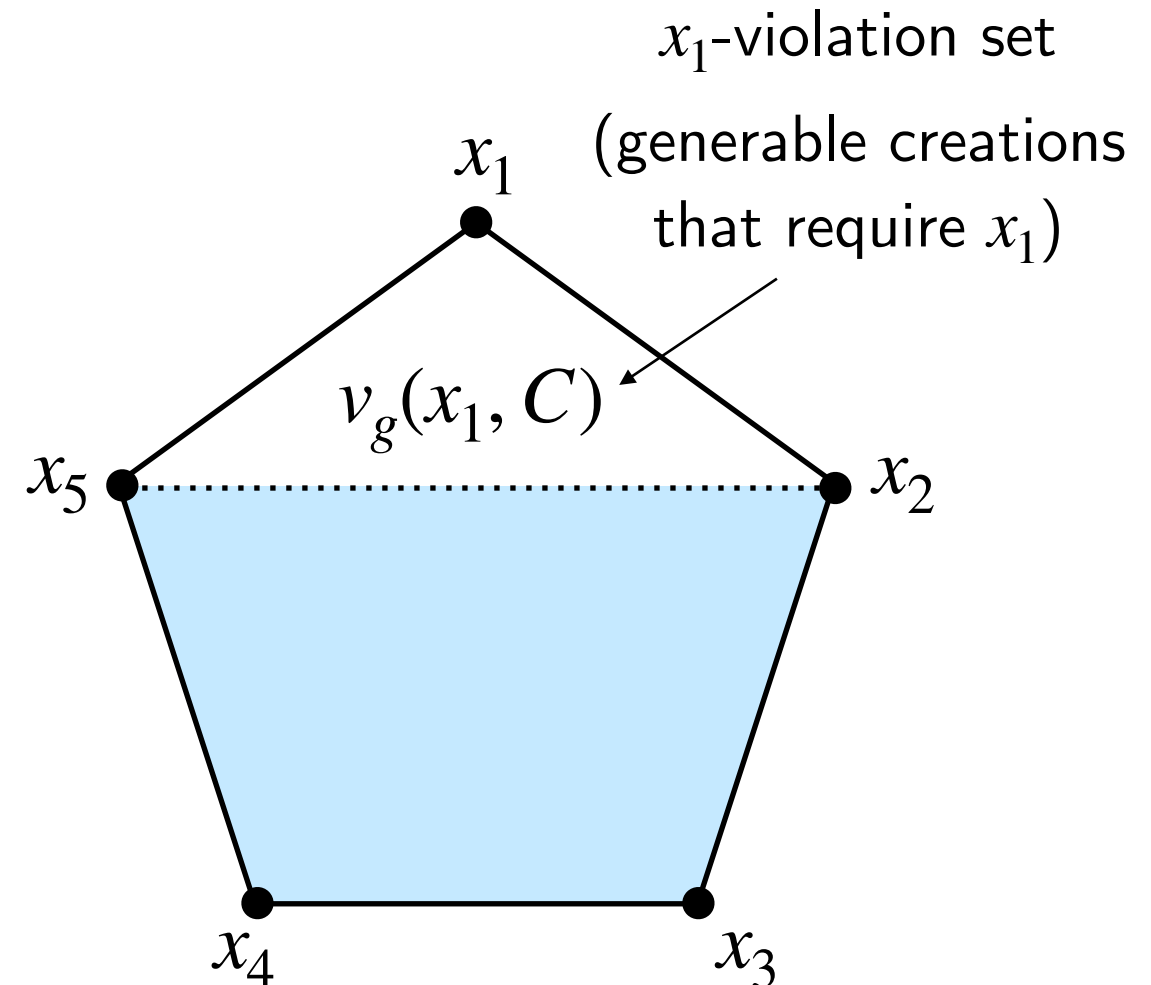
(generable creations that do not require x_1)

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1. The x -**permissible set** is $p_g(x, C) := g(C \setminus \{x\})$, i.e., all creations that are generable from $C \setminus \{x\}$
2. The x -**violation set** is $v_g(x, C) := g(C) - p_g(x, C)$, i.e., all creations that are generable from C but not from $C \setminus \{x\}$

Convex hull generator:



Permissible Generation

Definition:

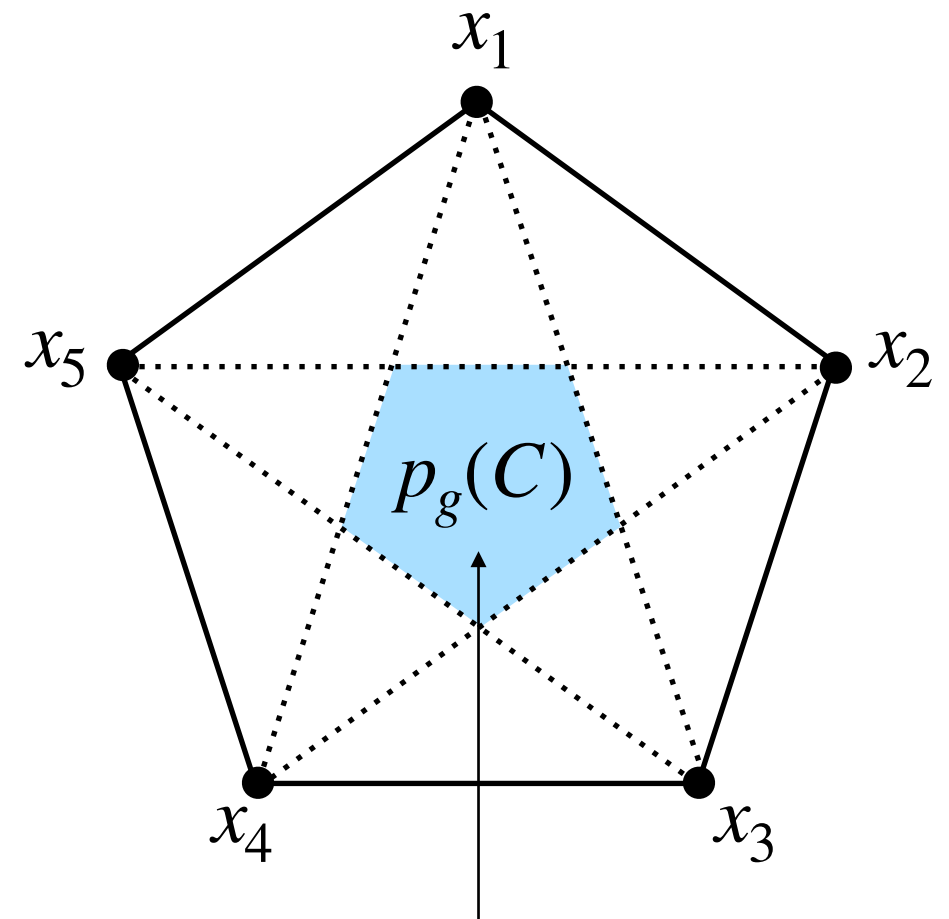
The **permissible set** is

$$p_g(C) := \bigcap_{x \in C} p_g(x, C)$$

The **violation set** is

$$v_g(C) := \bigcup_{x \in C} v_g(x, C)$$

Convex hull generator:



permissible set

(generable creations that do not critically rely on any $x \in C$)

Discussion

Q: What does it mean for an existing work to “have a claim”?

- The predecessor may be entitled to **attribution**.
- The predecessor may be entitled to **financial compensation or rent**.
- The downstream work may be **restricted or prohibited**.

Our framework is agnostic about which.

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Q: What does it mean for an existing work to “have a claim”?

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- The downstream work may be **restricted or prohibited**.

Our framework is agnostic about which.

Q: How would permissibility be evaluated in practice?

- Brute-force retraining of an algorithm is not feasible
- But machine unlearning provides methods to approximate the role of specific training points (Bourtoule et al., 2019; Xiong et al., 2024)
- Can also view permissibility as an evidentiary question and compare outputs across independently trained algorithms

Permissible Generation for Large Corpora

Preview

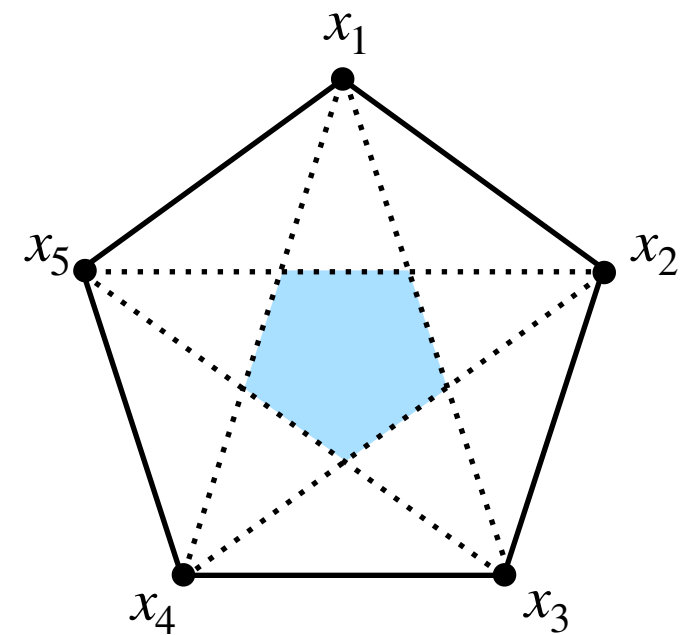
- Previously considered a fixed corpus
- Now ask how permissibility evolves as creative works accumulate
- Will individual creators retain rights over a nontrivial region of the creative space, or will eventually all generation be permissible?
- The answer depends on how the creative frontier expands
- Distinguish two cases:
 - Light-tailed innovation (e.g., Gaussian)
 - Heavy-tailed innovation (e.g., Pareto)

Permissible Ratio

Let $Vol(A)$ denote the Lebesgue measure (or *volume*) of the set $A \in \mathcal{C}$.

Definition: The **permissible ratio** is:

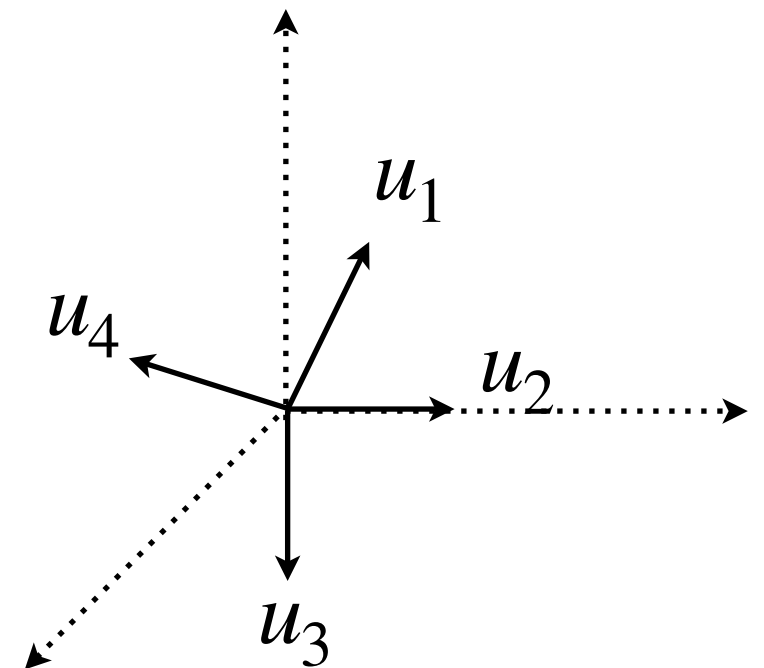
$$r_g(C) := \frac{Vol(p_g(C))}{Vol(g(C))}$$



i.e., the proportion (by volume) of generable creations that are permissible.

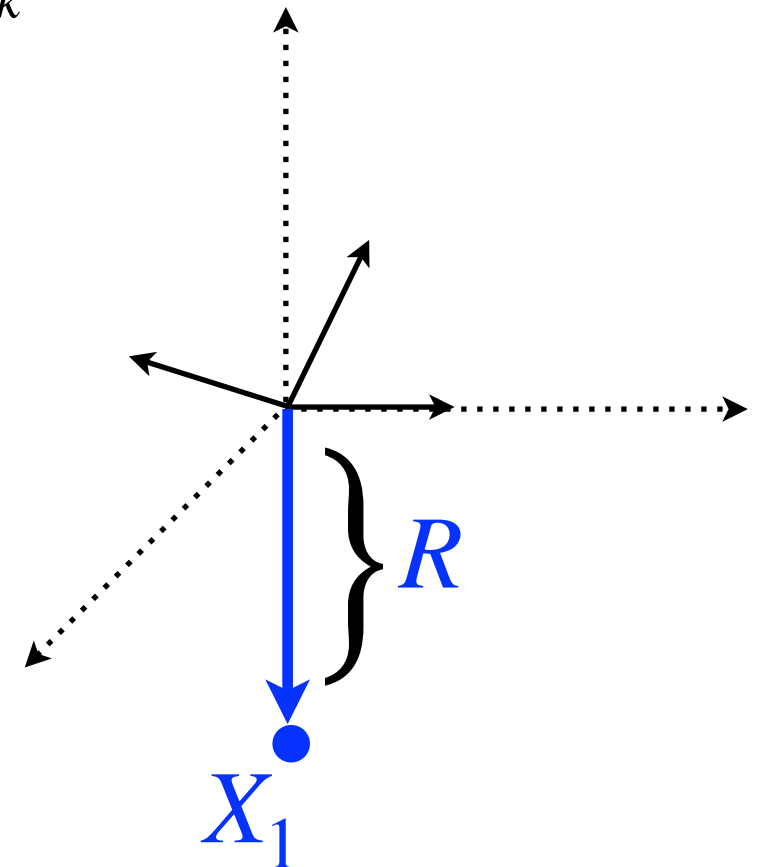
The Creative Process

- Ambient creative space: $\mathbb{R}^d$
- Fixed creative directions: $u_1, \dots, u_m \in \mathbb{R}^d$ (spanning $\mathbb{R}^d$)



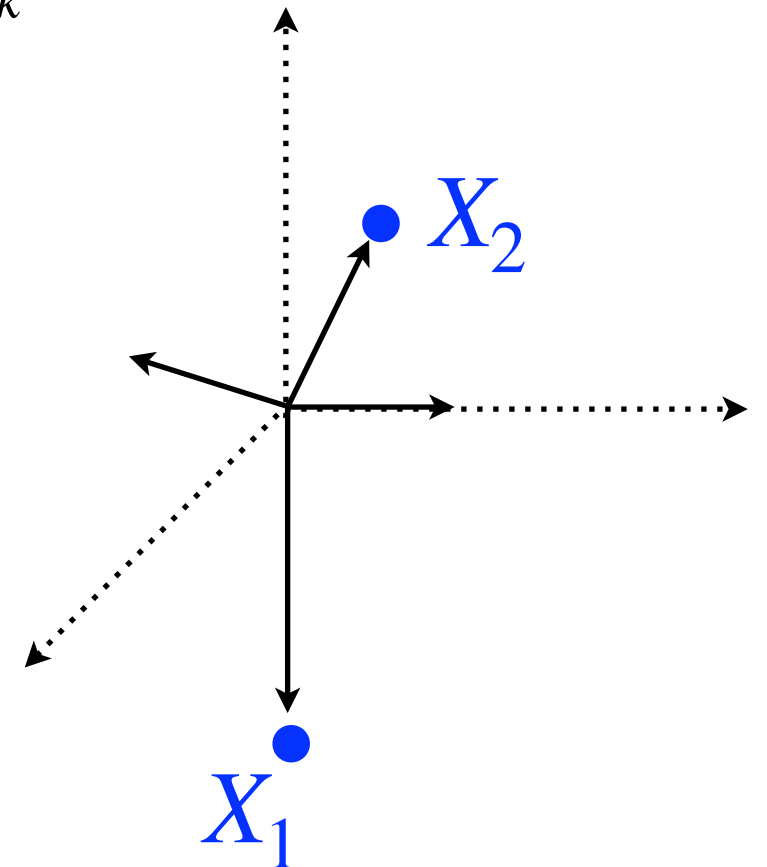
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- At each date n :
 - Choose a direction $k \in \{1, \dots, m\}$ (possibly given the history)
 - A magnitude $R \in \mathbb{R}_+$ is drawn from distribution F_k
 - The new creation is $X_n = Ru_k$



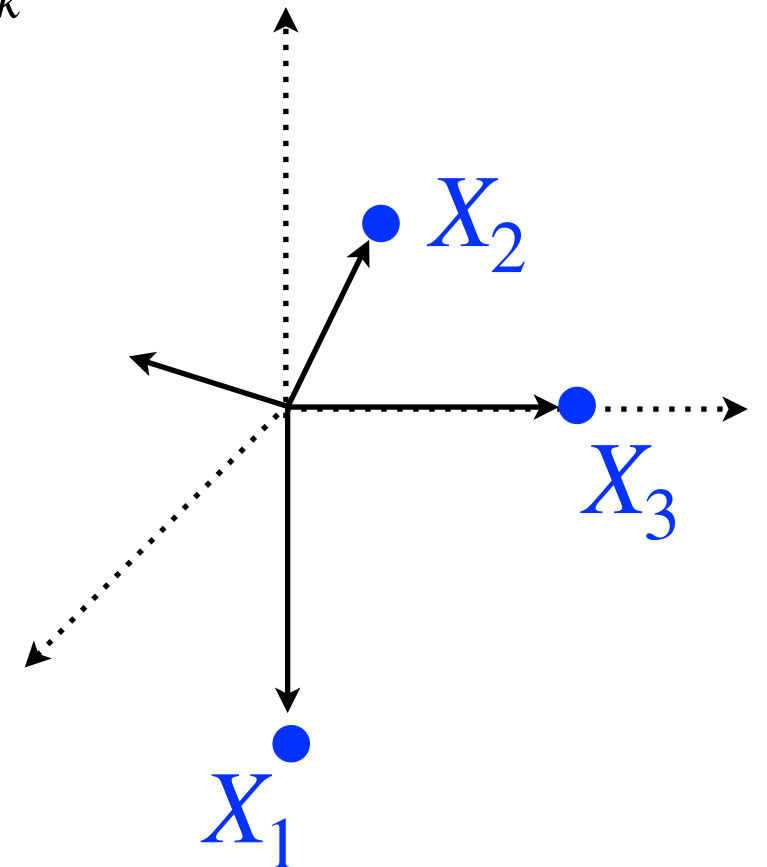
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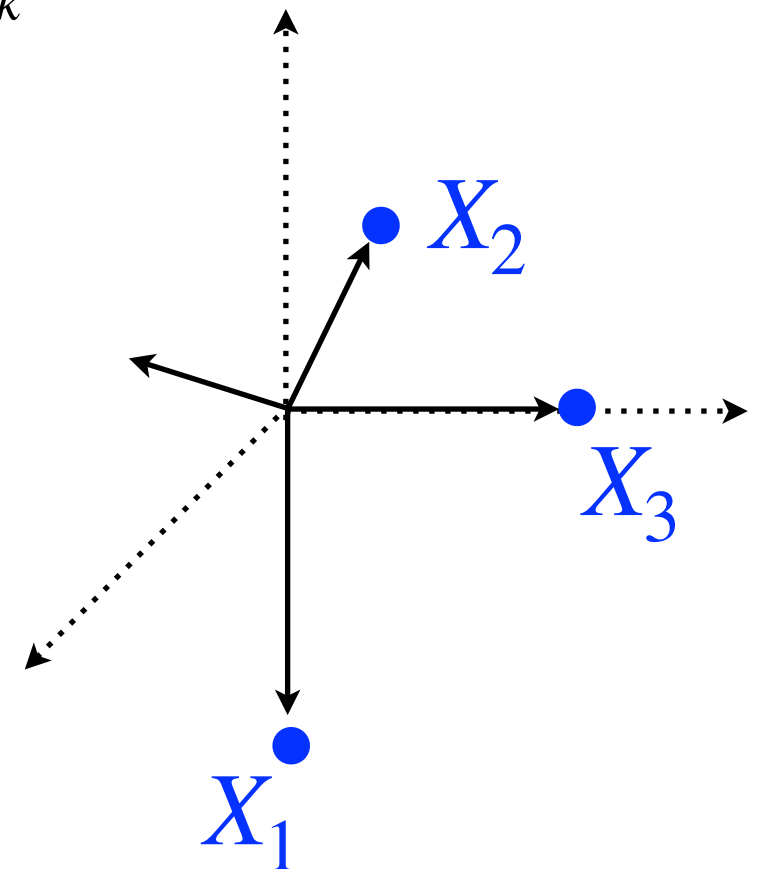
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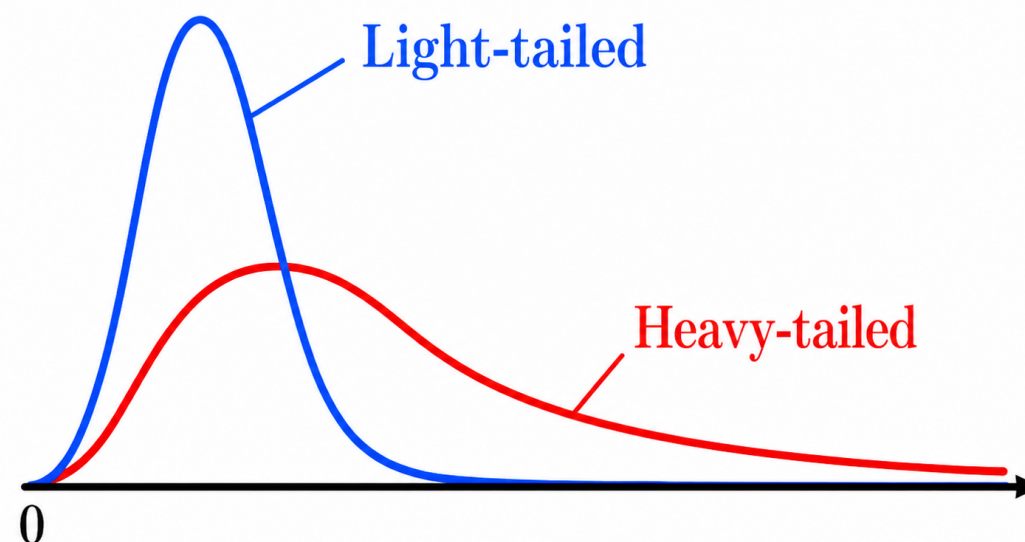
Question: What is $\lim_{n \rightarrow \infty} r_g(C_n)$, i.e. the fraction of generable works are eventually permissible?



Two Tail Regimes

Definition: Say that a distribution F with density f

- has **light tails** if its hazard rate $f(t)/(1 - F(t))$ is nondecreasing (e.g., **Gaussian**)
- has **heavy tails** if there is a constant α such that $(1 - F(bt))/(1 - F(t)) \rightarrow b^{-\alpha}$ for every $b > 0$ (e.g., **Pareto**)



Light-Tailed Result

Definition: A direction is persistent if it is sampled infinitely often, and the creative process is **expansive** if the persistent directions span $\mathbb{R}^d$.

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Definition: A direction is persistent if it is sampled infinitely often, and the creative process is **expansive** if the persistent directions span $\mathbb{R}^d$.

Proposition (informal): Fix a convex-valued and continuous generator g . If each F_k has light tails, and the process is expansive, then

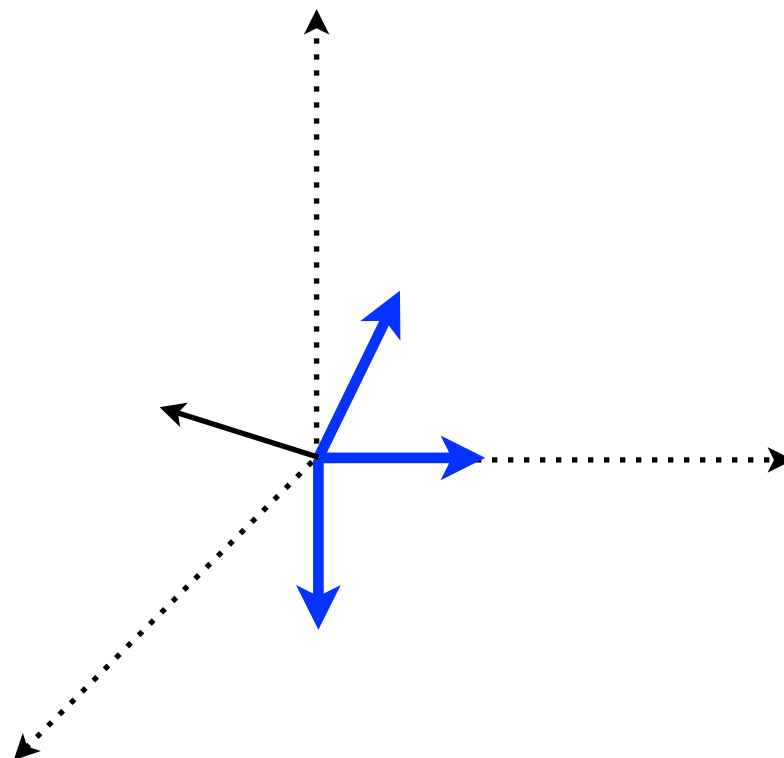
$$\lim_{n \rightarrow \infty} r_g(C_n) = 1 \quad \text{a.s.}$$

- In creative domains where innovation is largely incremental, the marginal contribution of any single work vanishes as the corpus expands.
- Generation is eventually unrestricted under our criterion.

Heavy-Tailed Result

Proposition (informal): Fix a convex-valued and bounded generator g . If F_k is heavy-tailed along a set of basis directions, which are each sampled a positive fraction of times, then:

$$\liminf_{n \rightarrow \infty} r_g(C_n) = 0 \text{ a.s.}$$



Heavy-Tailed Result

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- In creative domains characterized by rare but significant breakthroughs, protection for individual works retains its importance
- If the size of innovation is endogenous (in addition to direction), then we might expect incentives for creators to push towards this regime

Generalization to Protected Groups

- In practice, copyright disputes often arise at the level of collections rather than single works



Generalization to Protected Groups

- In practice, copyright disputes often arise at the level of collections rather than single works
- Can generalize our criterion to handle this
 - New primitive: a family of sets of works $\{A_1, \dots, A_n\}$ where each A_i is a **protected group**
 - Define the permissible set to be the intersection of the A_i -permissible sets (rather than the x_i -permissible sets)
- Our main result here: violations are **superadditive**
- If a firm that owns A_1 merges with a firm that owns A_2 , the permissible set of generations becomes smaller

Conclusion

- Have talked about
 - How AI clones will impact search over human candidates
 - How we might regulate AI creations
- Many interesting questions remain

Thank You

Discussion

Restrictiveness:

- Relative to the existing legal criteria, our proposed definition can be either more or less restrictive
- We view our criterion as a complement rather than a substitute for the existing criteria, could be that all should be simultaneously enforced

Implementation:

- Brute-force retraining of an algorithm with the counterfactual training corpus is generally not feasible
- But the emerging literature of machine unlearning develops methods to approximate exactly this (Bourtoule et al., 2019; Xiong et al., 2024)
- Moreover can compare outputs across independently trained algorithms for qualitative evidence

What is the effect of expanding the corpus?

Adding new works to the corpus:

- Expands the set of generable creations (monotonicity of g)
- Introduces new works whose rights must be respected (more constraints)

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Result: The permissible set is **monotone**:

$$C \subseteq D \implies p_g(C) \subseteq p_g(D)$$

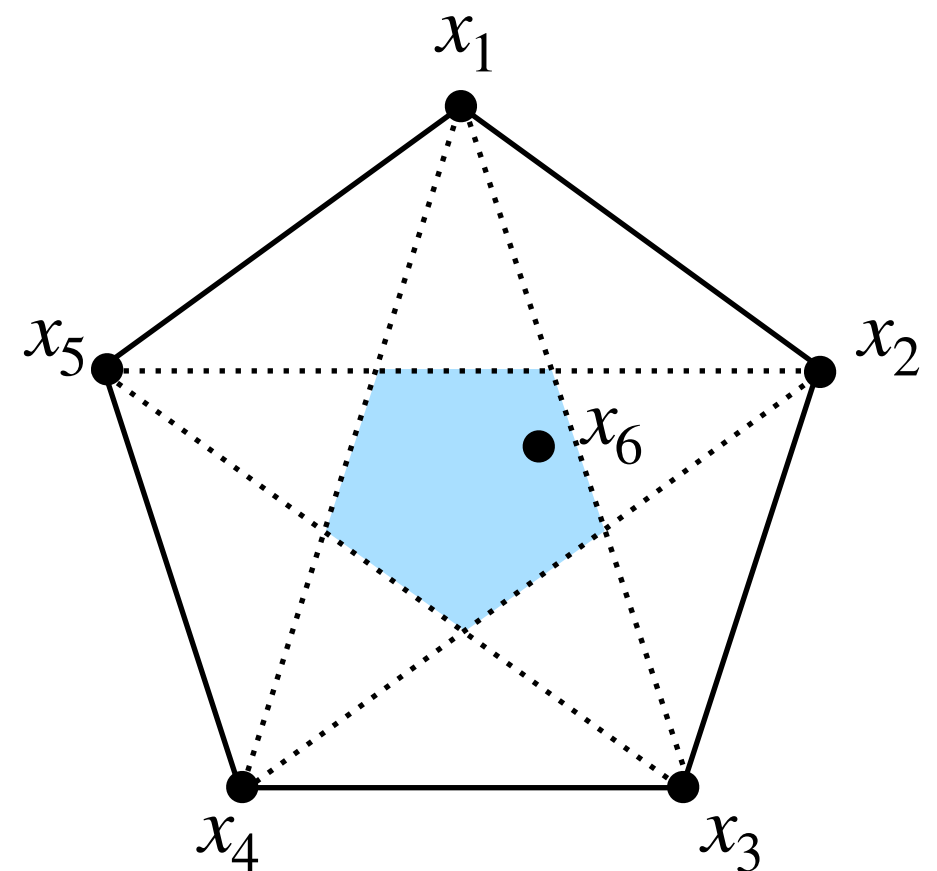
Adding new works to the corpus weakly expands the set of permissible generations.

What is the effect of expanding the corpus?

Result: Moreover, if $D = C \cup \{x\}$ then:

1. If x is permissible, the permissible set is unchanged:

$$x \in p_g(C) \implies p_g(C) = p_g(D)$$



What is the effect of expanding the corpus?

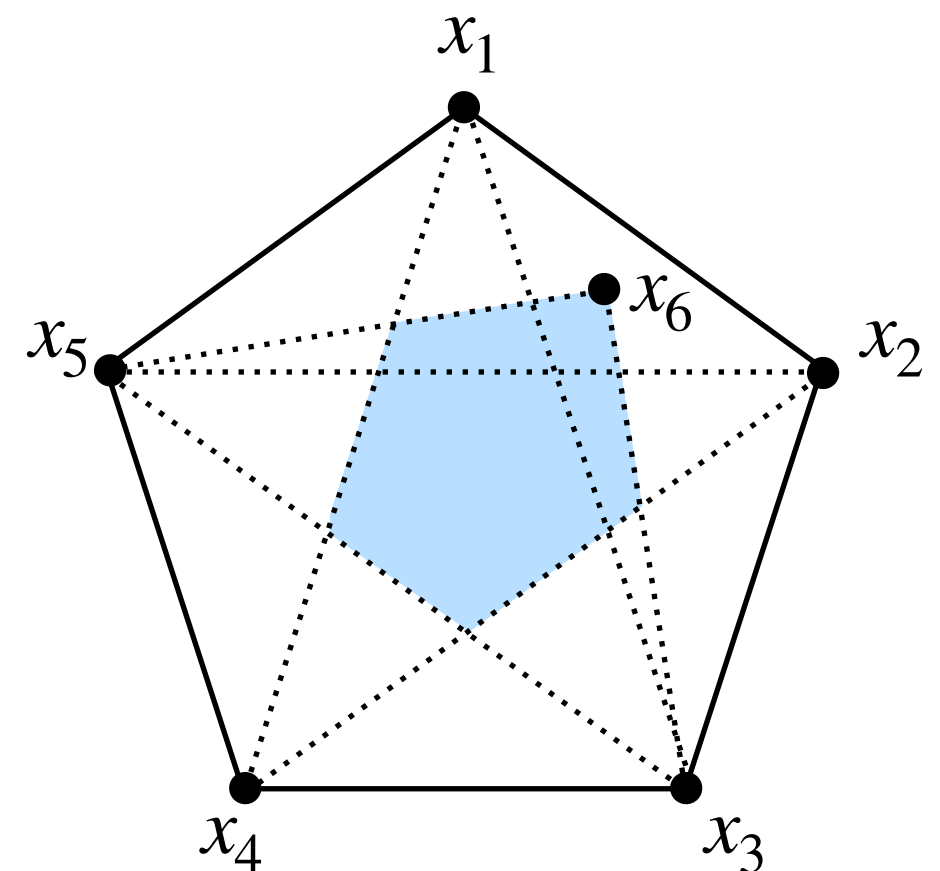
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2. If x is a violation, adding it strictly expands the permissible set:

$$x \in v_g(C) \implies p_g(C) \subsetneq p_g(D)$$



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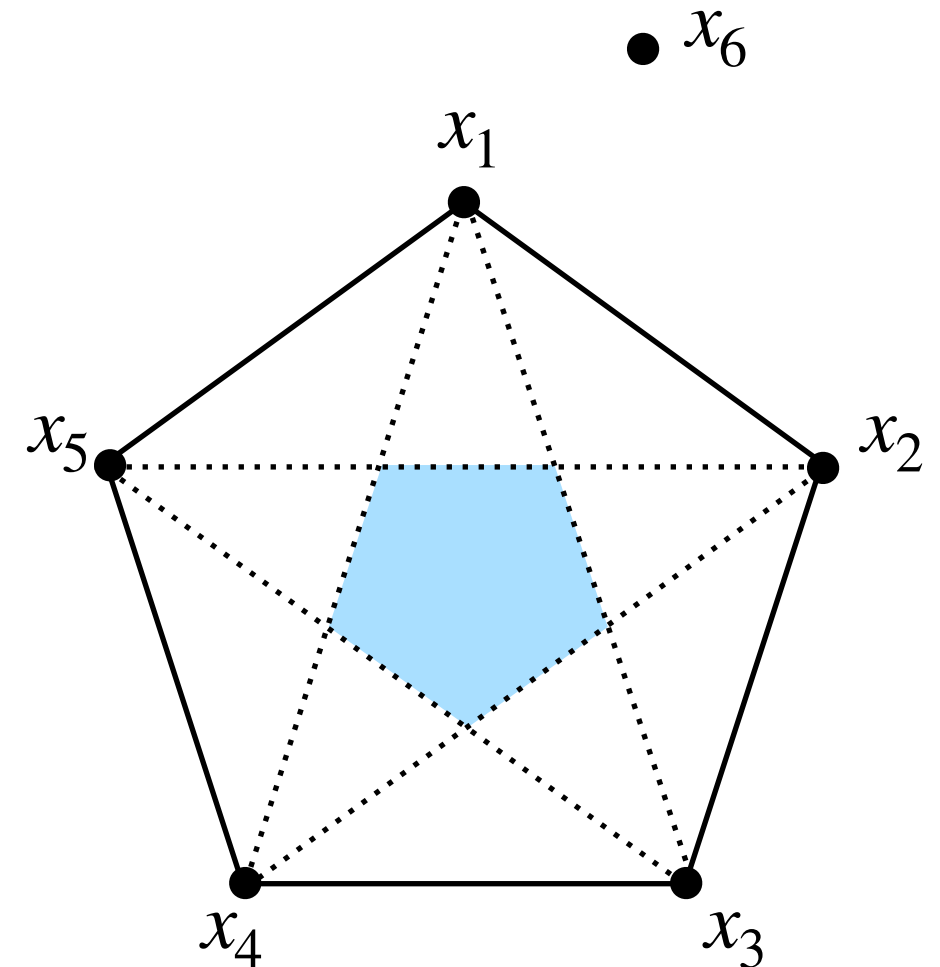
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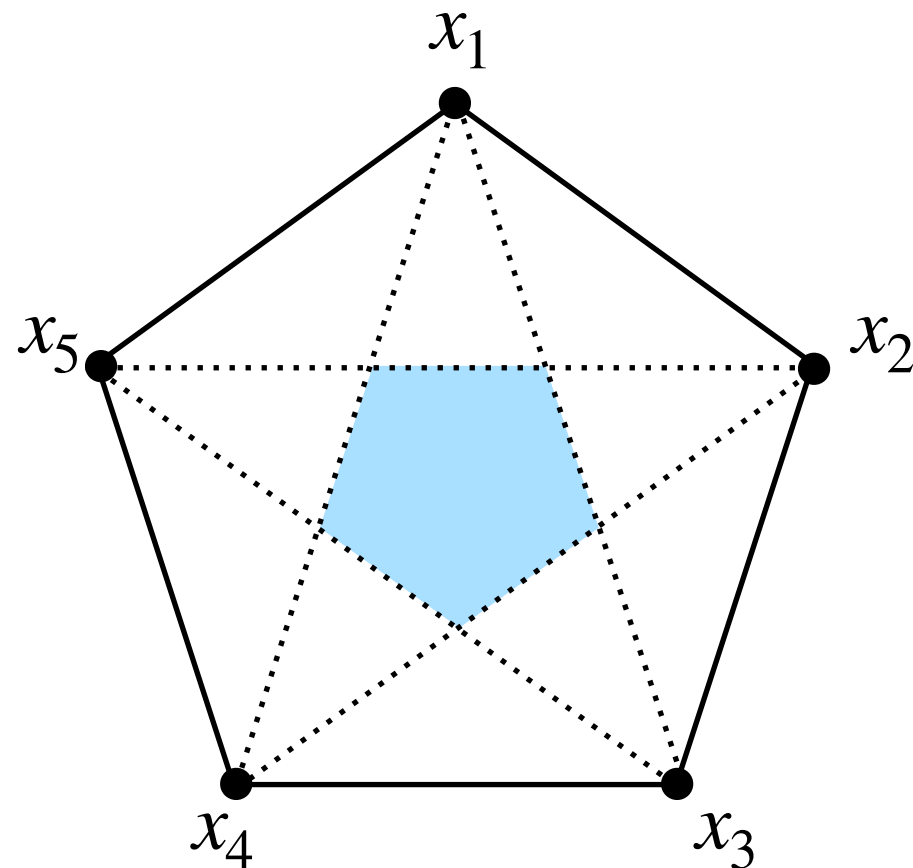
$$x \in v_g(C) \implies p_g(C) \subsetneq p_g(D)$$

3. If x is not generable (i.e., $x \notin g(C)$), then both equality and strict expansion are possible.



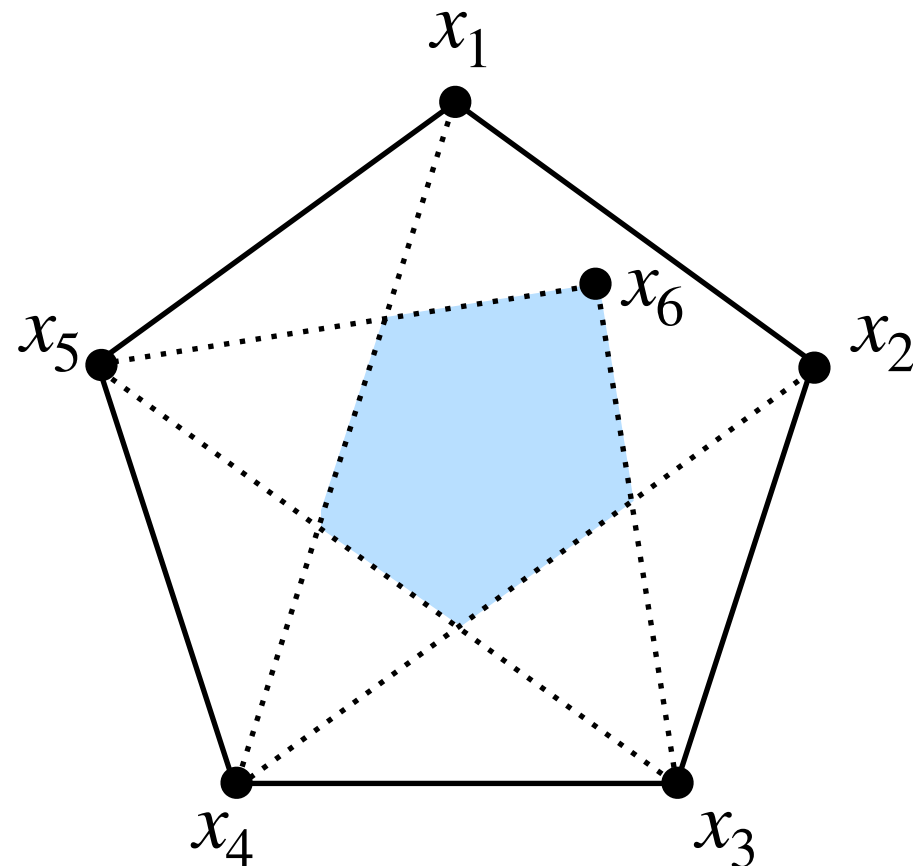
Long-Run Permissibility

Implication: If we continue to add new generations from $g(C)$, eventually everything will be permissible.



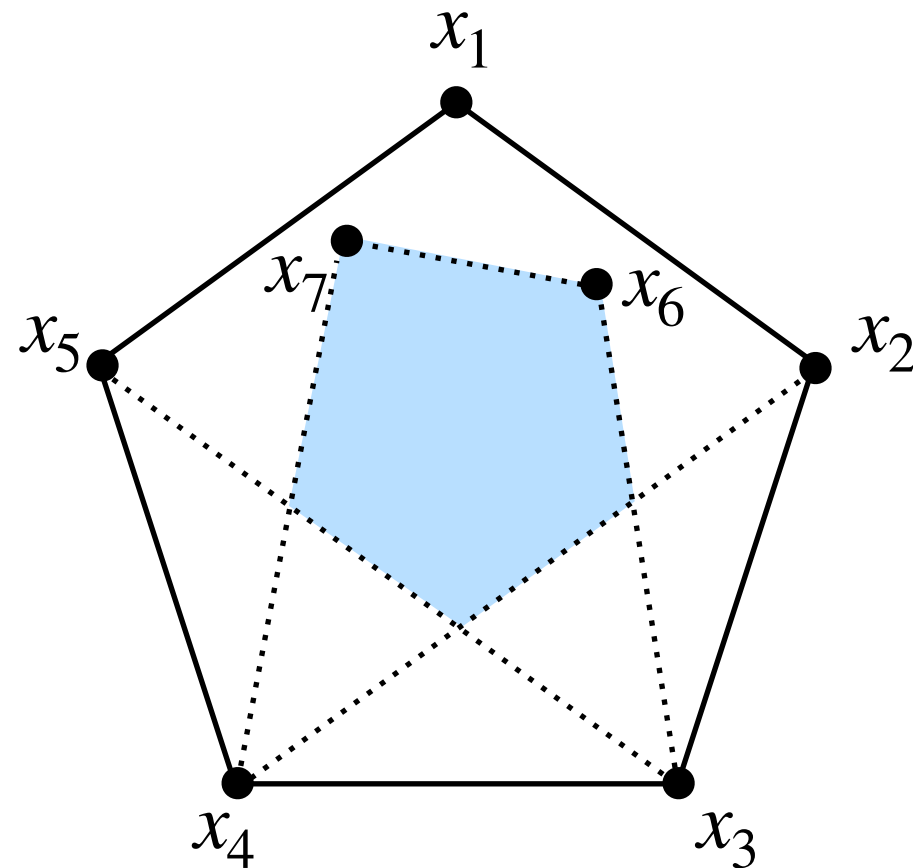
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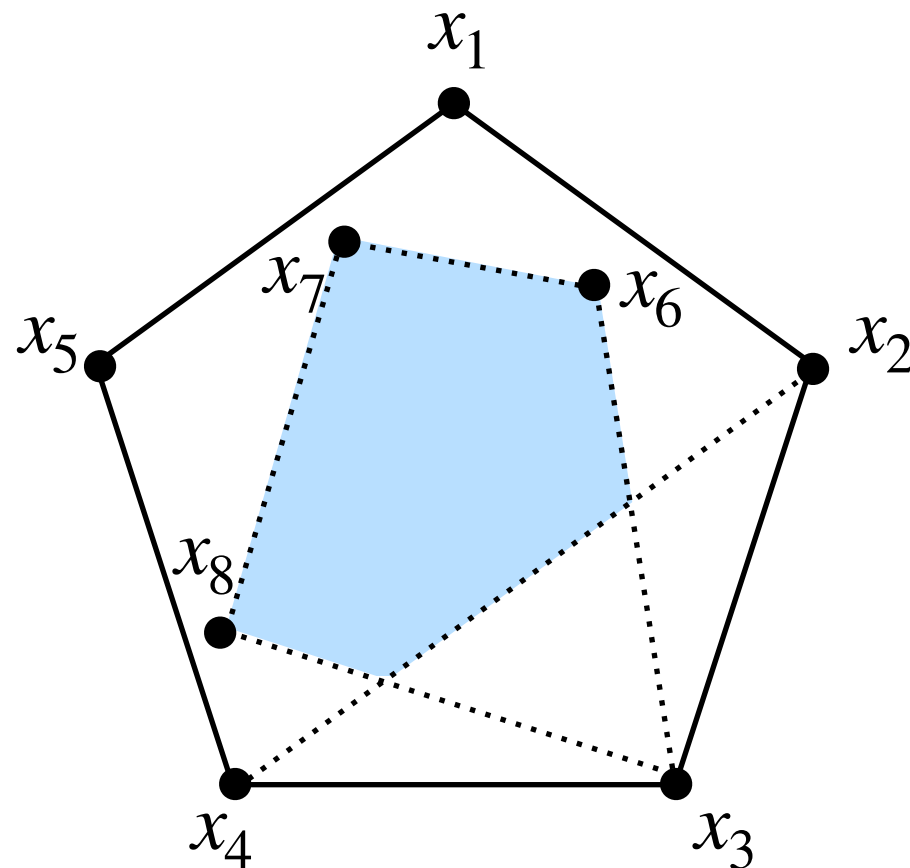
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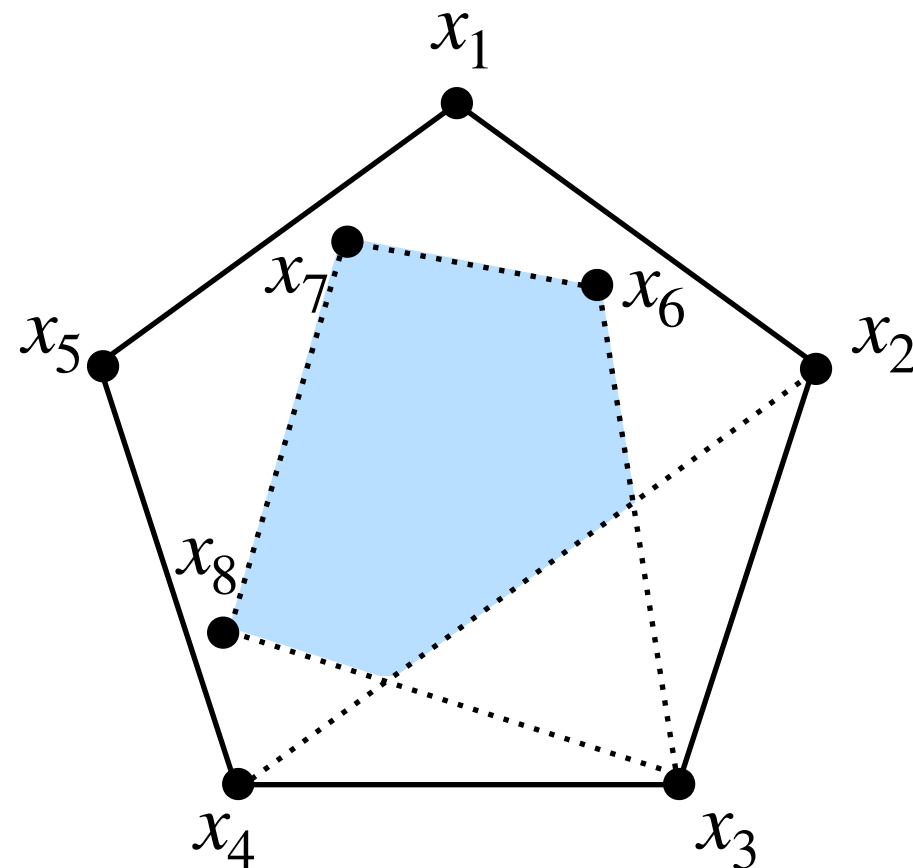
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But in addition, new **human** works are constantly being produced, which may be genuine innovations relative to the original corpus.

role of parametric assumptions

- personalities are uniformly distributed and AI errors follow a multivariate Gaussian distribution
- proof of **theorem 1** ($m_k^* = 2$ for large k) relies heavily on these assumptions, especially the rate of convergence lemmas
- but key statistical phenomenon is more general: in high dimensions,
 - random points isolate from one another
 - the distances between them concentrate around their expectations
- expect generalizations to other distributions (in particular, normal-normal)



two-sided matching

- this paper's model of search is one-sided
- abstracts from the question of whether Subject's best match (once identified) would agree to this pairing
- I do not pursue a complete matching model
- but in many such models, Subject's expected payoff would decrease in the number of candidates considered by Subject's potential partners
- this would further favor in-person search, where each individual searches over fewer candidates



increasingly accurate AI

- main results hold for any quality of AI approximation (i.e., any $\sigma > 0$).
- but would expect that as the AI representations become more accurate, AI-mediated search becomes more (relatively) attractive
- I show that for every fixed number of dimensions k ,

$$d_{k,\sigma}^{AI}(\infty) \rightarrow 0 \text{ as } \sigma \rightarrow 0$$

- i.e., Subject's expected distance to best match on the AI platform converges to zero as the AI's approximation errors vanish
- thus, while the value to increasing the size of the AI platform is capped, the value to increasing the accuracy of the AI clones is not



sequential search

have so far considered a static search problem

consider the following model of sequential search in discrete time

- at $t = 0$, Subject chooses between the in-person regime and AI regime
- in the **in-person regime**, observe X_t in each period $t \geq 1$ and decide whether to stop (and match with best so far) or sample again at cost c_{IP}
- in the **AI regime**,
 - pay an upfront cost κ to participate on the platform
 - in each $t \geq 1$, observe clone distance $\|Y_t - Y_0\|$ and decide whether to stop (and match with closest clone) or sample again at cost c_{AI}

result: Subject prefers the in-person regime when $\kappa > 2c_{IP}$ (i.e., the fixed cost of joining the platform exceeds the cost of meeting two people)



human-AI integration

- one might consider integrating the two approaches as follows:
 - the AI platform simulates interactions between Subject and n individuals using their AI clones
 - recommends to Subject those m candidates whose AI clones are closest to Subject's clone (previously just one)
 - Subject meets these individuals in person and matches to the most compatible among them
- this is **unlikely** to substantially improve Subject's payoffs



human-AI integration

- previously showed that the best match on the AI platform is (for large k) similar to a draw from a uniform distribution over the unit ball
- conjecture that the m best matches on the AI platform are also (for large k) similar to m draws from a uniform distribution on the unit ball
- thus choosing among the AI-sourced candidates is not substantially better than choosing from an equal number of randomly drawn candidates
- suggests that:
 - for any fixed platform entry cost, Subject should optimally choose not to participate on the platform when k is large
 - for large k , Subject should prefer searching over $m + 1$ people in person rather than selecting from m people sourced by the AI platform



intuition

- recall that the selected match i minimizes $\|Y_i^g - Y_0^g\|$
- in a large population, this is an individual for whom that distance is very small
- which group is that extreme draw more likely to come from?
- the smallest distance to any group- g individual, i.e.,

$$M_n^g = \min_{1 \leq i \leq n} \|Y_i^g - Y_0^g\|$$

is asymptotically distributed like an exponential random variable with rate proportional to $f_k^g(0)$ (the density of $Y_i^g - Y_0^g$ at the zero vector)

- so group g 's probability of “winning” is asymptotically determined by the relative size of its exponential rate, and thus by the size of $f_k^g(0)$
- finally, lower noise variance leads to a more sharply peaked density at zero, and hence a higher probability of being chosen

other discussion/extensions (see paper)

- role of parametric assumptions ➡
- two-sided matching ➡
- increasingly accurate AI ➡
- sequential search ➡
- human-AI integrated search ➡

conclusion

- many papers have compared human and AI evaluation on tasks that are hard for both humans and machines
- but there are important evaluations that are intrinsically easier for people
 - no machine or expert knows better than ourselves whose company we enjoy
- in such settings, human evaluation is necessarily more accurate than the AI evaluation, but also more costly
- can automated search over a sufficiently large number of candidates compensate for the AI's errors during evaluation?
- this paper suggests that the answer depends on the complexity of the target: when sufficiently many attributes are relevant, then the answer is **no**

Extended Model

- In practice there is heterogeneity in the AI's ability to represent different individuals
- Many reasons for this
 - different quantities of personal data to share with AI (e.g., because of variation in social media usage)
 - different willingness to share it (e.g., because of variation in privacy preferences)
- Next suppose the population is split into a “data-rich” and “data-poor” group

Extended Model

- Let $g \in \{R, P\}$ denote the group (data-rich or data-poor)
- Groups are identical in characteristics: each X_i^g independently drawn from a uniform distribution on the unit ball
- Individuals i from group g have AI representations

$$Y_i^g = X_i^g + \varepsilon_i^g, \quad \varepsilon_i^g \sim \mathcal{N}(0, \sigma_g^2 I_k)$$

where $\sigma_R^2 < \sigma_P^2$ (lower error for data-rich group)

- The AI platform searches over n candidates from each group

Extended Model

Proposition: in the limit as the population grows large, the probability that the searcher's match is from the data-rich group **exceeds $1/2$**

- Although the two groups have identical compatibility with the searcher, a data-rich individual is more likely to be identified as the best match

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Proposition: in the limit as the number of dimensions and population size both grow large, the probability that the searcher's match is from the data-rich group **converges to 1**